

Australian/New Zealand Standard™

Scaffolding

Part 2: Couplers and accessories



AS/NZS 1576.2:2016

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Australian/New Zealand Standard™

Scaffolding

Part 2: Couplers and accessories

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PREFACE

This Standard was prepared by the Standards Australia Committee BD-036, Scaffolding, to supersede AS/NZS 1576.2:2009.

The objective of this Standard is to provide manufacturers and suppliers of couplers and accessories with requirements that will ensure the proper operation of couplers and accessories for light, medium, heavy and special duty scaffolding.

Tubes, couplers and accessories that have been tested to AS/NZS 1576.2:2009, *Scaffolding*, Part 2: *Couplers and accessories*, are deemed to be compliant provided they have been tested and documented in accordance with the relevant Standard.

This edition incorporates the following major changes to the 2009 edition:

- (a) Couplers that meet the performance requirements of EN 74-1, *Couplers, spigot pins and baseplates for use in falsework and scaffolds*, Part 1: *Couplers for tubes—Requirements and test procedures*, for similar type couplers are deemed to comply with this Standard without the need for type testing.
- (b) Introduction of performance requirements for aluminium adjustable baseplates and aluminium adjustable swivel baseplates.
- (c) The scope of the vertical load test in Appendix H has been extended to cover both aluminium and steel baseplates.

The terms ‘normative’ and ‘informative’ have been used in this Standard to define the application of the appendix to which they apply. A ‘normative’ appendix is an integral part of a Standard, whereas an ‘informative’ appendix is only for information and guidance.

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Australian/New Zealand Standard
Scaffolding

Part 2: Couplers and accessories

SECTION 1 SCOPE AND GENERAL

1.1 SCOPE

This Standard specifies requirements for couplers and accessories for scaffolding complying with AS/NZS 1576.3 and AS/NZS 1576.6.

NOTE: Guidance on means for demonstrating conformity with this Standard is given in Appendix P.

1.2 NORMATIVE REFERENCES

The following are the normative documents referenced in this Standard:

NOTE: Documents referenced for informative purposes are listed in the Bibliography.

AS

- 1111 ISO metric hexagon bolts and screws (all parts)
- 1112 ISO metric hexagon nuts (all parts)
- 1237 Plain washers for metric bolts, screws and nuts for general purposes (all parts)
- 1444 Wrought alloy steels—Standard, hardenability (H) series and hardened and tempered to designated mechanical properties
- 1450 Steel tubes for mechanical purposes
- 1874 Aluminium and aluminium alloys—Ingots and castings

AS/NZS

- 1163 Cold-formed structural steel hollow sections
- 1554 Structural steel welding
- 1554.1 Part 1: Welding of steel structures
- 1576 Scaffolding
- 1576.1 Part 1: General requirements
- 1576.3 Part 3: Prefabricated and tube-and-coupler scaffolding
- 1576.6 Part 6: Metal tube-and-coupler scaffolding—Deemed to comply with AS/NZS 1576.3
- 1594 Hot-rolled steel flat products
- 1665 Welding of aluminium structures
- 1734 Aluminium and aluminium alloys—Flat sheet, coiled sheet and plate
- 1866 Aluminium and aluminium alloys—Extruded rod, bar, solid and hollow shapes
- 2451 Bolts, screws and nuts with British Standard Whitworth threads (rationalized series)

AS/NZS

3678 Structural steel—Hot-rolled plates, floorplates and slabs

3679 Structural steel (all parts)

EN

74 Couplers, spigot pins and baseplates for use in falsework and scaffolds

74-1 Part 1: Couplers for tubes—Requirements and test procedures

1.3 DEFINITIONS

For the purpose of this Standard, the definitions given in AS/NZS 1576.1 and those below apply.

1.3.1 Adjustable baseplate

A baseplate embodying an adjustable leg.

1.3.2 Adjustable castor

A castor incorporating a height-adjusting device that fits inside the standard or vertical member of a scaffold.

1.3.3 Adjustable leg

A threaded bar or tube with nut, designed to fit inside a standard to support the load from a standard and which, in conjunction with a baseplate, U-head or castor, is for levelling purposes.

1.3.4 Adjustable swivelling baseplate

An adjustable baseplate incorporating a hinge between the baseplate and the leg.

1.3.5 Adjustable U-head

A U-head embodying an adjustable leg.

NOTE: An adjustable U-head is also known as an adjustable fork head.

1.3.6 Baseplate

A plate to distribute the load from a vertical loadbearing member to the supporting structure.

1.3.7 Castor assembly

A wheel rotating on an axle fixed into a housing capable of being connected to the lower end of a standard for the purpose of supporting and moving a scaffold.

NOTE: The housing may be fixed or may be capable of swivelling to allow the scaffold to be moved horizontally in any direction.

1.3.8 Check coupler

A right angle coupler, swivel coupler or parallel coupler that is fixed hard against a loadbearing coupler to increase the slip resistance along the tube.

1.3.9 Coupler

A fitting that joins two tubes.

1.3.10 End-to-end coupler

Sleeve coupler or expanding joint pin that connects and aligns the tube end-to-end.

1.3.11 Flange clamp

A clamp that connects a tube to the flange of a structural member.

NOTE: Flange clamps may be rigid or swivelling.

1.3.12 Joint pin

An internal end-to-end coupler for joining two tubes.

1.3.13 Parallel coupler

A coupler for making a lap or spliced joint between two tubes.

1.3.14 Pintle

A projection, at the top of the castor, used to locate the vertical members of the mobile scaffold frame.

1.3.15 Putlog blade

A fitting that is fixed to the end of a putlog permitting it to be located and supported in a joint of a wall.

1.3.16 Putlog coupler

A coupler for fixing a putlog to a ledger.

1.3.17 Right-angle coupler

A non-swivel loadbearing coupler, other than a putlog coupler, that connects two tubes at right angles.

1.3.18 Screw-tightened coupler

A coupler in which the clamping force on the tubes is provided by tightening the flaps around the tube by means of a bolt and nut.

1.3.19 Sleeve coupler

An external end-to-end coupler for joining two tubes.

1.3.20 Swivel coupler

A coupler for connecting two tubes at any angle.

1.3.21 Swivelling baseplate

A baseplate that incorporates a hinge and is capable of being connected to a standard, adjustable leg or adjustable baseplate.

1.3.22 U-head

A U-shaped housing to support and locate a standard over a bearer or a bearer over a standard.

NOTE: A U-head may be fixed or adjustable.

1.3.23 Wedge-tightened coupler

A coupler in which the clamping force on the tubes is provided by tightening the flaps around the tube by means of a wedge hammered home.

1.4 ALTERNATIVE DESIGN METHODS AND MATERIALS

Alternative design methods, materials and the like, which are not specified in this Standard, may be used provided it can be demonstrated that the resulting couplers or accessories satisfy the requirements of this Standard.

Service durability shall be a consideration in any assessment of new materials.

1.5 MARKING

Each coupler and accessory, excluding baseplates, shall be marked with the manufacturer's or supplier's name or trademark, and a code traceable to a manufacturing batch.

The marking shall be impressed or embossed on the flap or body of the coupler or accessory and shall be visually legible after the protective coating has been applied. The height of the characters shall be at least 4 mm and the impressed or embossed dimension shall be at least 0.2 mm.

1.6 CONFORMITY WITH EUROPEAN STANDARD

Couplers that have been tested and have demonstrated conformity with the performance requirements of EN 74-1:2005 for Class B right-angle couplers and Class A swivel couplers, where these requirements are equivalent to the performance requirements of this Standard, are deemed to comply with this Standard and do not require further testing. Such couplers may not comply with dimensional requirements of this Standard, in which case they shall be assessed against the requirements herein before they are deemed acceptable.

1.7 INFORMATION TO BE SUPPLIED

The following documented information shall be supplied with couplers and accessories, where applicable:

- (a) Tube dimensions and material for which the coupler has been designed.
- (b) Tightening procedure for conformity with the performance requirements of this Standard.
- (c) Maximum tightening torque or force, as appropriate, to avoid overstressing of the material of the coupler or accessory.
- (d) The maximum acceptable distortion of any part of the coupler before it is deemed not to satisfy the requirements of Section 4.
- (e) Servicing procedures for couplers and accessories that will ensure continuance of the performance requirements of Section 4.
- (f) Working load limit (WLL) of adjustable legs and adjustable baseplates.

SECTION 2 MATERIALS

2.1 PROPERTIES

Where the properties of any material are in doubt, the material shall be subjected to such test or tests necessary to determine the properties concerned.

2.2 STEEL

Steel couplers and accessories shall be manufactured from steel complying with AS/NZS 1163, AS 1444, AS 1450, AS/NZS 1594, AS/NZS 3678 or AS/NZS 3679, as appropriate.

Welded components shall comply with the requirements of AS/NZS 1554.1.

2.3 ALUMINIUM

2.3.1 Cast components

Cast aluminium alloy components shall comply with AS 1874.

The 700 series alloy shall not be used.

NOTE: Pressure die-casting is not recommended as a method of manufacture for primary loadbearing members. Where pressure die-casting is used, the component design should include sufficient redundancy to compensate for potential reduction in capacity due to the possibility of high internal porosity within the component.

2.3.2 Wrought components

Wrought aluminium alloy components shall comply with AS/NZS 1866 for extrusions or AS/NZS 1734 for sheet and plate.

The 2000 and 7000 series alloys shall not be used.

2.3.3 Welded components

Welded components shall comply with the requirements of AS/NZS 1665.

SECTION 3 DESIGN REQUIREMENTS

3.1 GENERAL

Couplers and accessories shall be designed and manufactured so that they will firmly fix or fit their intended tubes, even after repeated use.

3.2 COUPLERS

3.2.1 General requirements

All couplers shall comply with Clauses 3.2.2 and 3.2.3.

3.2.2 Configuration

The following requirements apply to coupler design and construction:

- (a) When assembled, but before being tightened, no part shall become inadvertently detached.
- (b) When tightened, the coupler shall not be inadvertently released;
- (c) Where the thinnest section of the body or flap of any steel coupler is less than 3.2 mm thick, that section shall be hot-dip galvanized.
- (d) Threaded blind holes shall not be used. The design of a coupler shall be such that the engaged length of any threaded components can be readily determined when assembled.
- (e) Hinge pins shall be steel of not less than 8 mm diameter and shall be adequately riveted over or otherwise suitably prevented from becoming displaced.
- (f) Where fitted, closing flaps on all couplers shall have their range of movement restricted such that, when fixed to a vertical member with the other flap fully open, the open flap is horizontal (within $\pm 5^\circ$) to provide support for a horizontal scaffold tube during installation.
- (g) Where couplers incorporate nuts and bolts or screws to effect tightening, they shall comply with the following requirements:
 - (i) A tightening bolt or screw shall comply with AS 1111 (series) or AS/NZS 2451 and shall have a shank not less than 11 mm diameter and threaded overall diameter not less than 12 mm.
 - (ii) Nuts shall comply with the material requirements of AS 1112 or AS/NZS 2451 and shall be not less than 11 mm thick. Bolts shall project past the nut to a maximum of 25 mm. Nuts shall be made captive on the bolt.
 - (iii) Washers shall comply with the AS 1237 series and shall be not less than 2.5 mm thick. Washers shall be fitted beneath the nut, or be integral with the nut.
 - (iv) A screw-tightened coupler shall be designed such that, when tightened to its highest specified torque (see Clause 3.2.3), the tightening bolt or screw shall have not less than 2 mm of travel remaining.

NOTE: This requirement should produce a minimum gap of 2 mm between the flap and the coupler body when the coupler is fixed to a tube with a diameter at the lower manufacturing tolerance.
- (h) Except when the geometry of the fitting is such that, when tightened, the bolt is positively held in position, a nib or recess shall be provided on the coupler flap to prevent the nut from being displaced from the coupler flap when being tightened.

- (i) Neither the tightening wedge nor the end of the bolt or screw shall directly touch the tube in order to tighten the coupler unless the contact of wedge or bolt along the tube is greater than 18 mm in all cases. If necessary, a permanently attached spacer shall be incorporated to protect the tube.
- (j) It shall not be possible to accidentally disengage the tightening wedge during work on the scaffold. The tightening wedge shall be captive to the coupler.
- (k) When used as designed, couplers shall not distort the scaffold tube (see Clause 4.1.1).

3.2.3 Tightening requirements

The following apply to tightening of couplers:

- (a) For a screw-tightened coupler, a torque of 54 Nm shall be sufficient for the coupler to satisfy the requirements of Section 4.
- (b) For a screw-tightened coupler to achieve a slip resistance performance greater than the requirements of Section 4, a higher tightening torque than 54 Nm may be specified, provided the maximum stress in the screwed component does not exceed 70% of the minimum yield stress of the material.
- (c) The method of tightening the wedge shall enable the coupler to achieve its design performance in accordance with Section 4. In the absence of information on the method being provided by the supplier or manufacturer, the coupler shall be tightened by striking the wedge twice firmly with a 500 g hammer swung through a distance of between 300 mm and 320 mm.

3.2.4 Robustness

Couplers shall be capable of being tightened to a minimum of 80 Nm around the tube without any part of the specified coupler or its components showing any sign of permanent distortion. After release of the torque and removal of the coupler, the coupler shall be in a serviceable condition.

Where a tightening torque higher than 80 Nm is specified to achieve added slip resistance, the robustness requirement shall apply for a torque 50% above the higher specified tightening torque.

3.2.5 Swivel couplers

In addition to the requirements of Clauses 3.2.2 and 3.2.3, swivel couplers shall have a steel swivel pin not less than 200 mm² in area with a minimum tensile strength of 380 MPa. The heads of the pin shall be well formed and free from any riveting defects. The swivelling faces shall be smooth and in contact. This requirement shall not prohibit a metal washer being interposed between the faces of the coupler. Where a metal washer is inserted, the coupler shall not be less efficient.

NOTE: Swivel couplers should not be used to transfer primary loads unless specified in the installation design.

3.2.6 End-to-end couplers

3.2.6.1 General

End-to-end couplers shall comply with the following:

- (a) They shall be self-centring.
- (b) They shall ensure that the tubes are co-axial.
- (c) They shall not be used to resist tensile forces between the joined tubes.
- (d) They shall not be used in the assembly of a truss unless made redundant by an appropriately designed tension splice.

- (e) A positive stop shall ensure that the fitting accepts an equal length of each tube. The stop shall provide an even surface and effective bearing area for the end of the tube, which shall have been cut squarely.
- (f) Under normal loadings, the bearing surface shall not distort the end of any tube with which it is in contact.

3.2.6.2 Internal types

Each spigot of an internal type (joint pin) shall extend not less than 73 mm on the internal surface of each tube they connect to. The spigot shall be capable of being expanded within a tube while it is engaged.

3.2.6.3 External types

External types (sleeve coupler) shall have a length of contact not less than 73 mm on each tube they connect, and shall provide an effective grip between the tube and the coupler. They shall have not less than two tightening bolts, one on each half of the length of the fitting. Washers shall be required where the coupler does not provide a full bearing surface for the nut.

The requirements of Clause 3.2.2(g)(i) may be varied for end-to-end couplers to enable bolts less than 12 mm overall diameter to be used, provided the nuts comply with Clause 3.2.2(g)(ii).

3.3 ACCESSORIES

3.3.1 Baseplate

A baseplate shall have a minimum area of 20 000 mm², with a minimum length or width of 120 mm. A shank, of length not less than 50 mm and diameter not less than 16 mm, or equivalent cross-sectional area, shall be secured to the centre of the upper surface. The shank shall be a loose fit to allow insertion of its full length inside the tube. The upper surface of the baseplate shall provide a smooth and even bearing for the end of the tube and shall be such as not to damage or distort the tube or itself be damaged or distorted under WLL.

Steel used for baseplates shall be of a minimum thickness of 6 mm and of a minimum yield strength of 230 MPa.

Aluminium used for baseplates shall be a minimum thickness of 10 mm and a minimum yield strength of 215 MPa.

Other materials may be used provided they have been designed to achieve equivalent strength and rigidity.

3.3.2 Swivelling baseplate

A swivelling baseplate shall have the same area and thickness and material composition as a baseplate.

3.3.3 Adjustable leg or adjustable baseplate

The following apply to an adjustable leg or adjustable baseplate:

- (a) It shall have a shank with a maximum extension adjustment not exceeding 600 mm.
- (b) When fully extended, the length of the shank that remains in the scaffold standard shall be the greater of $L/4$ and 150 mm, where L is the overall length of the shank.

Where the difference between the outside diameter of the shank and the inside diameter of the scaffold standard is greater than 7 mm, a washer, spacer or other device shall be fixed at the top of the shank to reduce the difference in diameter to 7 mm or less. The adjusting nut shall incorporate a spigot, socket or other device to accommodate and positively locate the end of the scaffold standard.

Where an existing adjustable leg or adjustable baseplate design has a length of shank which does not comply with the length specified above and which remains in the tube, it may continue to be used provided it can be demonstrated that the product satisfies the load performance of the system in which it is to be used.

NOTE: The adjusting nut spigot, socket or other device should have a maximum diametral clearance of not more than 3 mm and a nominal depth of 3 mm for positive location (see Figure 3.1).

- (c) The adjusting nut shall provide a flat, even bearing surface across the full contact face of the bearing standard tube allowing for any eccentricity resulting from any difference between the spindle diameter and the inside diameter of the standard tube.
- (d) For tubular stem adjustable leg or adjustable baseplate, the minimum wall thickness shall be a nominal 5 mm.

NOTES:

- 1 The hollow stem of a steel adjustable leg or adjustable baseplate should be hot-dip galvanized internally and externally.
 - 2 A drainage hole, not less than 12 mm in diameter, should be placed near the centre of a steel baseplate that has a tubular spigot, shank or stem fixed or attached to the baseplate.
- (e) The threaded portion of the adjusting nut shall be not less than 0.875 times the nominal thread diameter (see d in Figure 3.1).
 - (f) An adjusting swivelling baseplate shall have—
 - (i) the same minimum area and thickness of a baseplate; and
 - (ii) the baseplate shall be captive to the shank.

3.3.4 Castors

Castors shall comply with the following requirements:

- (a) All castors shall be clearly marked with the WLL as assembled.
- (b) The plain pintle of a castor shall be not less than 150 mm long.
- (c) The adjustable leg of an adjustable castor shall comply with Clause 3.3.3 as applicable.
- (d) Castors fitted with plain pintles shall include means to retain them to the end of the tube so that they do not dislodge from the tube when it is lifted off the supporting surface.

NOTE: Most existing adjustable castors include this facility.

- (e) Castors fitted with plain pintles shall include the facility to attach them to the end of the tube so that they do not dislodge from the tube when it is lifted off the running surface.
- (f) The diameter of the wheel shall be not less than 125 mm, and the eccentricity of the axle relative to the pintle shall not exceed 65 mm.
- (g) A castor shall be provided with an effective wheel brake mechanism that requires a positive action to release and cannot be accidentally released due to movement on the mobile scaffold. The effectiveness of the wheel brake shall be such that the mobile scaffold is held stationary, on a supporting surface with a gradient of 5°, when the brakes are applied to all but one of its castors.
- (h) Wheels shall be either plain or fitted with resilient tyres. Tyres shall be of adequate section and properly fixed on to the wheel centre. Pneumatic tyre wheels shall not be used.

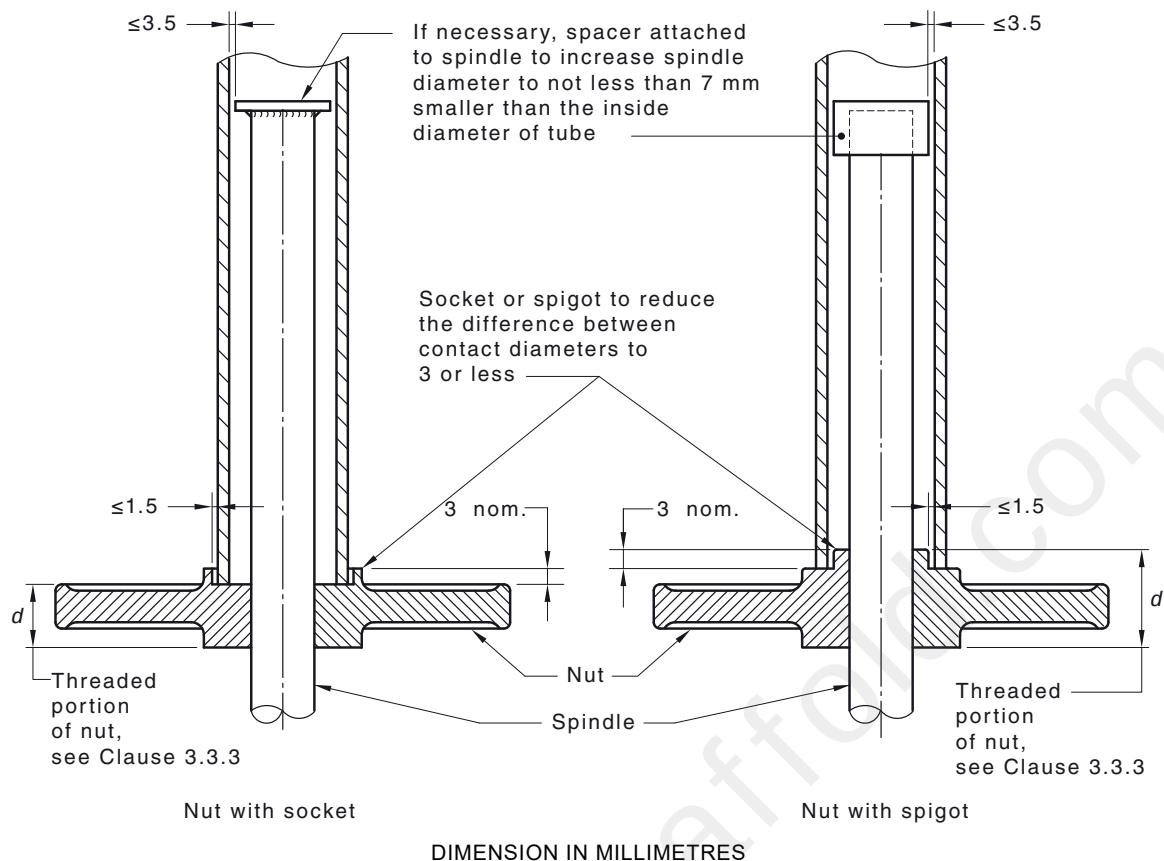


FIGURE 3.1 ADJUSTABLE LEG SHANK

3.3.5 U-head

A U-head shall be nominally concentric with the connecting member and shall have a recess not less than 75 mm deep.

The U-shape recess shall be formed from steel plate not less than 75 mm wide × 6 mm thick.

3.3.6 Putlog blade

In addition to the requirements of Clauses 3.2.2 and 3.2.3, the following requirements shall apply to putlog blades:

- The fitting shall contain a suitable stop to prevent the end of the tube from encroaching into the blade area.
- The blade section shall be constructed of minimum 75 mm long × 45 mm wide × 6 mm thick steel with the stop so positioned that the blade has at least a 75 mm long bearing surface. The bearing surface may be tapered at the end to a thickness of not less than 5 mm. It shall be not less than 60 mm long before tapering commences.

3.3.7 Spanner

The spanner for tightening couplers shall have a handle not less than 200 mm nor more than 250 mm long.

The socket of socket type spanners shall be 5 mm deeper than the bolt projection specified in Clause 3.2.2(g)(ii).

3.3.8 Flange clamp

The thinnest section, bolt, nut and pin of a flange clamp shall comply with the requirements of Clauses 3.2.2 and 3.2.3.

SECTION 4 PERFORMANCE REQUIREMENTS

4.1 GENERAL

All couplers and accessories from a sample submitted for testing shall pass the relevant performance requirements of this Section. If any coupler or accessory from the sample submitted for testing fails to meet the acceptance criteria, then the whole batch shall be rejected.

NOTE: A sample is the group of couplers or accessories selected for testing from a manufacturing batch.

4.2 COUPLERS

4.2.1 Sampling requirement

A minimum of six couplers shall be tested for each of the relevant test Appendices.

4.2.2 Acceptance criteria

4.2.2.1 General

Each coupler type shall be tested in accordance to the relevant test procedure laid out in the appendices.

After each test—

- (a) each coupler shall be capable of being reassembled onto the scaffold tube for which it has been designed;
- (b) swivel couplers and swivelling flange clamps shall freely swivel through 360°; and
- (c) distortion of the tube or tubes shall not exceed the following:

$$\text{Local distortion} = \left(\frac{\text{Maximum diameter of distorted cross-section}}{\text{Diameter of undistorted adjacent cross-section}} \right) \leq 1.03$$

4.2.2.2 Right-angle coupler

When tested in accordance with Appendix A, a right-angle coupler, in each position of the coupler, shall not slip more than 6 mm at its contact face.

When tested in accordance with Appendix B, the right-angle coupler, in each position of the coupler, shall not show a vertical rotation greater than 125 mm as indicated by the pointer.

4.2.2.3 Swivel coupler

When tested in accordance with Appendix A, a swivel coupler, in each position of the coupler, shall not slip more than 6 mm at its contact face.

When tested in accordance with Appendix B, the swivel coupler, in each position of the coupler, shall not show a vertical plane rotation greater than 175 mm as indicated by the pointer.

4.2.2.4 Putlog coupler

When tested in accordance with Appendix C, a putlog coupler shall not have a visible gap between the tubes, and the tube that was loaded shall not be able to be moved by hand in the coupler, longitudinally or rotationally.

When tested in accordance with Appendix D, the putlog coupler shall not slip on the tube.

4.2.2.5 *End-to-end coupler*

An end-to-end coupler shall be subjected to a bending test about the weakest bending axis or plane of the coupler.

NOTE: Preliminary testing may be required to establish the weakest bending axis or plane.

When tested in accordance with Appendix E, an external type coupler shall reach 15 kN and not deflect more than 12 mm under the test load. An internal type coupler shall reach 7.5 kN and not deflect more than 12 mm under the test load. The deflection shall be measured at the joint between the tubes.

4.2.2.6 *Parallel coupler splicing test*

When tested in accordance with Appendix F, parallel couplers or swivel couplers used to splice tubes shall not slip more than 6 mm at their contact face.

4.2.2.7 *Check coupler*

When tested in accordance with Appendix G, the load rating of a loadbearing coupler backed by a check coupler shall be limited to half the load at which a slip of not more than 6 mm along the tube occurs.

The check coupler shall not have a working slip resistance of less than 9 kN.

4.2.2.8 *Used couplers*

Used couplers, if requiring re-assessment, shall be tested to demonstrate conformity with the performance requirements of this Section.

4.3 ADJUSTABLE LEG, ADJUSTABLE BASEPLATE OR ADJUSTABLE SWIVEL BASEPLATE

4.3.1 **General**

The test component shall be tested as follows:

- (a) Components shall be tested in accordance to Appendix H.
- (b) Components that have passed Appendix H shall be tested to failure at which load the component is unable to support an increase in load or there is a slip between the thread of the nut and the thread of the stem. This load shall be the ultimate load.
- (c) A component whose ultimate test load deviates from the mean result by more than 10% shall be discarded and an additional component shall be tested.
- (d) The ultimate test results shall satisfy the following:

$$\begin{aligned} \text{WLL} &\geq 30 \text{ kN (for steel components), or} \\ &15 \text{ kN (for aluminium components)} \end{aligned}$$

where

$$\text{WLL} = (\sum P_u / n) / 2.5$$

P_u = individual ultimate load, in kilonewtons

n = number of tests

NOTE: The stated WLL of adjustable legs, adjustable baseplates or adjustable swivel baseplates may be further reduced as governed by the scaffold system design.

4.3.2 **Sampling requirement**

A minimum of three components shall be tested.

4.3.3 Acceptance criteria

When tested in accordance with Appendix H at a load of 60 kN for steel components and 30 kN for aluminium components, an adjustable leg or adjustable baseplate or adjustable swivel baseplate—

- (a) shall not show change in straightness of the shank exceeding 2 mm; and
- (b) the adjusting nut shall be able to turn by hand, for the full depth of the nut, downwards from the test position and to return upwards to the test position.

4.4 CASTOR WITH PLAIN PINTLE

4.4.1 Sampling requirement

A minimum of three castors shall be tested.

4.4.2 Acceptance criteria

When tested in accordance with Appendix I, a castor—

- (a) shall not show permanent set at a load of 1.5 times the rated load;
- (b) shall not show any sign of component separation or component failure at two times the rated load;
- (c) shall not collapse at a load three times the rated load; and
- (d) shall not show permanent set exceeding 3 mm regardless of load.

NOTE: Separation of components includes an increasing gap between the plates housing the castor rotation balls as well as release of any of the bearing components.

4.5 ADJUSTABLE CASTORS

4.5.1 Sampling requirement

A minimum of three castors shall be tested.

4.5.2 Acceptance criteria

When tested in accordance with Appendix J, an adjustable castor—

- (a) shall not show increasing gap between the plates housing the castor rotation balls as well as release of any of the balls;
- (b) shall not show separation of any components or collapse of the adjustable castor at two times the rated WLL;
- (c) shall not show a change in straightness exceeding 2 mm; and
- (d) the adjusting nut shall be able to turn by hand, for the full depth of the nut, downwards from the test position and return it to the test position.

4.6 PUTLOG BLADE

4.6.1 Sampling requirement

A minimum of six putlog blades shall be tested.

4.6.2 Acceptance criteria

When tested in accordance with Appendix K, a putlog blade shall not show permanent set.

4.7 RIGID OR SWIVELLING FLANGE CLAMP

4.7.1 General

Flange clamps are used in pairs to connect a metal scaffold tube to parallel flange or tapered flange beams.

Flange clamps shall comply with Clause 4.2.2.1.

4.7.2 Sampling requirement

A minimum of six pairs of flange clamps shall be tested for each of the test Appendices L, M and N.

4.7.3 Acceptance criteria

Each flange clamp type shall be tested in accordance to the relevant test procedure laid out in the appendices. After each test—

- (a) each flange clamp type shall be capable of being reassembled onto the scaffold tube for which it has been designed;
- (b) swivelling flange clamps shall freely swivel through 360°; and
- (c) distortion of the tube or tubes shall not exceed the following:

$$\text{Local distortion} = \left(\frac{\text{Maximum diameter of distorted cross-section}}{\text{Diameter of undistorted adjacent cross-section}} \right) \leq 1.03$$

When tested in accordance with Appendix L—

- (i) the flange clamps shall not show any permanent set; and
- (ii) provided the flange clamps pass the 30 kN load test, the WLL shall be not greater than 15 kN for the pair of flange clamps.

When tested in accordance with Appendix M, the clamped tube shall not have a slippage of more than 6 mm in relation to the flange of the beam.

When tested in accordance with Appendix N, the flange clamps shall not have a slippage of more than 6 mm along the flange of the beam.

SECTION 5 TESTED ITEMS

Any coupler or accessory used in a test required by this Standard shall not be used subsequently in a scaffold, regardless of whether permanent set or distortion was not evident.

Provided no tested item failed any test, untested items from the submitted batch subsequently may be used in scaffolding.

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APPENDIX A

SLIP TEST FOR RIGHT-ANGLE AND SWIVEL COUPLERS

(Normative)

A1 SCOPE

This Appendix sets out the method for determining the slip of right-angle and swivel couplers on metal scaffold tube.

A2 PRINCIPLE

The coupler is fixed to two scaffold tubes at right angles. A predetermined force is applied to one of the tubes. The amount of slippage of the coupler on the other tube is measured.

A3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure A1. The frame shall be a stiff structure that does not deflect under test loads.
- (b) Scaffold tube for which the coupler has been designed.
- (c) Torque wrench for a screw-tightened coupler.
- (d) The tightening device for a wedge-tightened coupler in accordance with Clause 3.2.3(b).
- (e) Timer.

A4 TEST COMPONENTS

New or as-new hot-dip galvanized steel scaffold tubes with nominal wall thickness of 4.0 mm shall be used. New or as-new aluminium scaffold tubes with nominal wall thickness of 4.47 mm shall be used. When tested, the tubes shall have a surface condition similar to that of a new or as-new tube. Where the surface has been marked by slippage of a coupler during a prior test, a fresh length shall be used.

A5 PROCEDURE

The procedure shall be as follows:

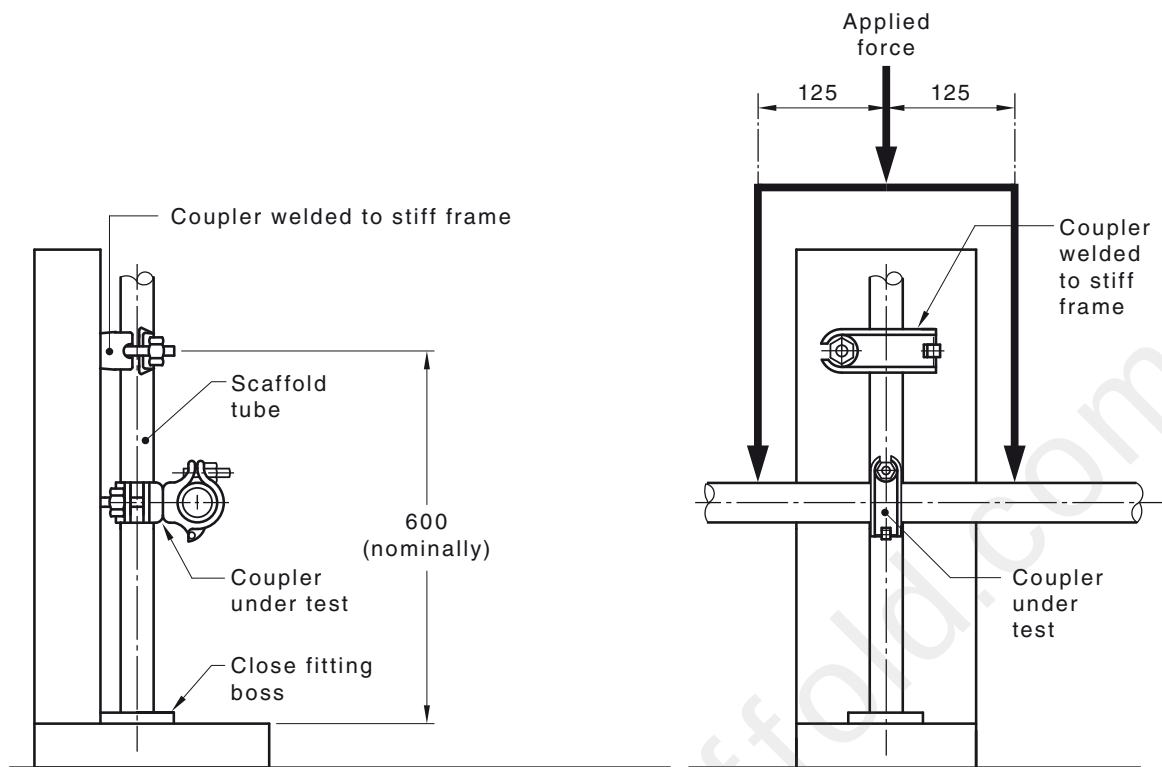
- (a) Fix the coupler on the scaffold tube as shown in Figure A1. Where the coupler incorporates a tightening bolt it shall be in the uppermost position. Nuts shall be free running on the bolt. Where the coupler uses a wedge for tightening, the wedge shall be in the orientation in which it was designed to be used.
- (b) For screw-tightened couplers, lubricate with light machine oil the thread of the bolt.
- (c) Tighten the coupler to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten the coupler to the requirements of Clause 3.2.3 and wait a minimum of 10 min before applying the initial settling in force.
- (e) Subject the coupler to a vertical force of 6 kN applied progressively over a period of 15 s and hold it for 1 min -0 , $+15$ s. The force shall be applied in the same manner as the test force (the force is to permit the coupler to bed in). All readings shall be taken after this force has been released.

- (f) Re-tighten the coupler to the requirements of Clause 3.2.3.
- (g) Scribe a mark on the vertical tube immediately above the coupler.
- (h) Apply a vertical force of 12.5 kN progressively over a period of 30 s and hold it for 1 min –0, +15 s. Where a screw-tightened coupler is tested for a higher slip resistance, the test load shall be twice the intended WLL.
- (i) Apply the test force progressively over a longer period than Step (h) above, by linear interpolation and hold for 1 min –0, +15 s.
- (j) Remove the force, scribe a new mark on the tube immediately above the coupler and measure and record the amount of slip.
- (k) Invert the coupler.
- (l) Repeat Steps (b) to (i).
- (m) Invert the coupler so that the part of the coupler, which was previously attached to the horizontal tube, is now attached to the vertical tube.
- (n) Repeat Steps (b) to (i).
- (o) Invert the coupler.
- (p) Repeat Steps (b) to (i).

A6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of couplers tested.
- (b) The number of couplers that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4.
- (c) The test force applied to the scaffold tube.
- (d) For each couple position—
 - (i) the load at commencement of slip; and
 - (ii) the amount of slip, in millimetres.
- (e) A reference to this test method, i.e. AS/NZS 1576.2, Appendix A.



NOTE: Coupler shown is swivel coupler.

DIMENSIONS IN MILLIMETRES

FIGURE A1 SLIP TEST APPARATUS FOR RIGHT-ANGLE AND SWIVEL COUPLERS

APPENDIX B

DISTORTION TEST FOR RIGHT-ANGLE AND SWIVEL COUPLERS

(Normative)

B1 SCOPE

This Appendix sets out the method for determining the distortion of right-angle and swivel couplers on metal scaffold tube, where the coupler has passed the slip test (see Appendix A).

B2 PRINCIPLE

The coupler is fixed to two scaffold tubes at right angles and is supported at the sides and back to prevent it moving down the tube. A predetermined force is applied to one of the tubes and the vertical plane distortion is measured after the force is removed.

B3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure B1. The pointer shall be extended to cover the angular displacement of the pointer to measure the vertical distortion against a scale in a vertical plane. The frame shall be a stiff structure that does not deflect under test loads.
- (b) Scaffold tubes for which the coupler has been designed.
- (c) Torque wrench for a screw-tightened coupler.
- (d) The tightening device for a wedge-tightened coupler in accordance with Clause 3.2.3(b).
- (e) Timer.

B4 TEST COMPONENTS

Test specimens, as specified in Paragraph A4, Appendix A (i.e., right-angle or swivel coupler), which have been previously tested in accordance with Appendix A and passed that test.

B5 PROCEDURE

The procedure shall be as follows:

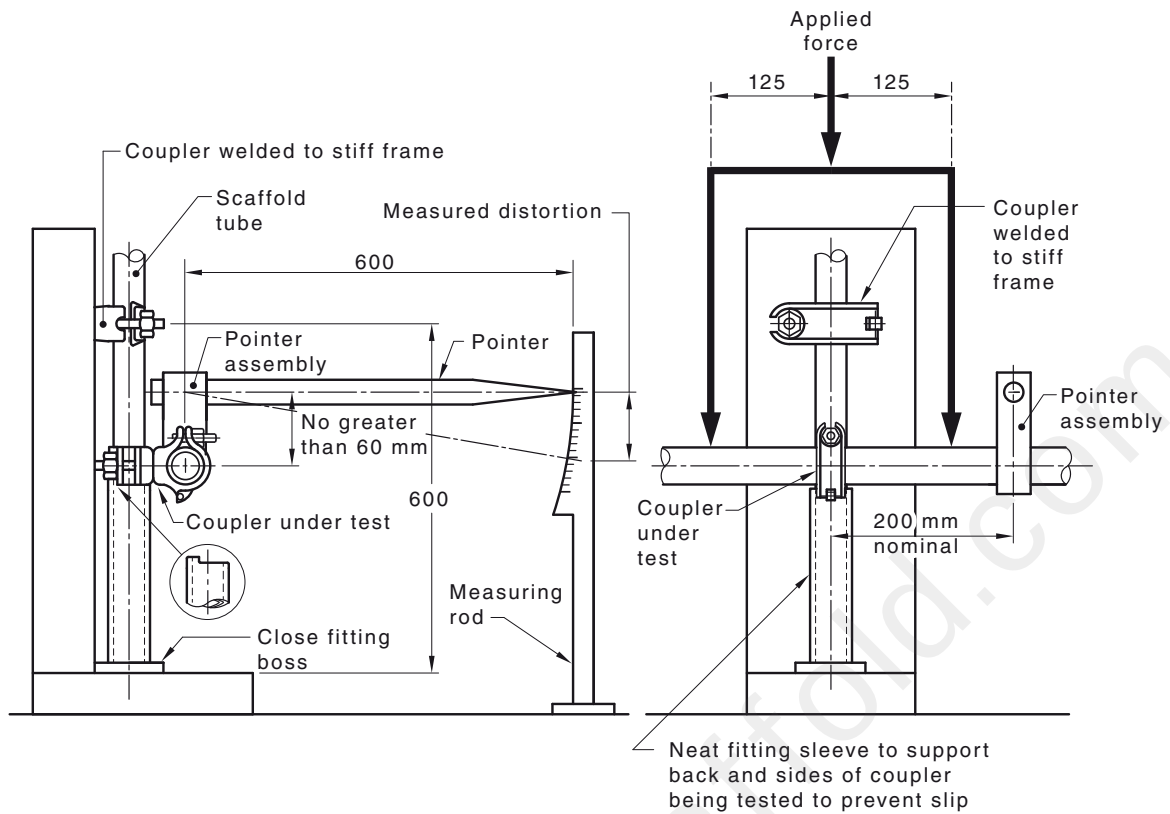
- (a) Fix the coupler on the scaffold tube as shown in Figure B1. Where the coupler incorporates a tightening bolt it shall be in the uppermost position. Nuts shall be free-running on the bolt. Where the coupler uses a wedge for tightening, the wedge shall be in the orientation in which it was designed to be used.
- (b) For screw-tightened couplers, lubricate the thread of the bolt with light machine oil.
- (c) Tighten the coupler to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten the coupler to the requirements of Clause 3.2.3 and wait a minimum of 10 min before applying the initial settling in force.

- (e) Subject the coupler to a vertical force of 6.0 kN progressively over a period of 15 s and hold it for 1 min –0, +15 s. The force shall be applied in the same manner as the test force (this force is to permit the coupler to bed in). All readings shall be taken after this force has been released.
- (f) Re-tighten the coupler to the requirements of Clause 3.2.3.
- (g) Read the pointer position on the measuring scale.
- (h) Apply the following force vertically and hold it for 1 min –0, +15 s once it has been reached:
 - (i) For a right-angle coupler 30 kN applied progressively over 90 s.
 - (ii) For a swivel coupler 15 kN applied progressively over 45 s.
- (i) Remove the force and measure and record the amount of vertical plane rotation.
NOTE: The pointer on the measuring rod measures this vertical plane rotation.
- (j) Invert the coupler so that the tightening bolt or tightening wedge referred to in Step (b) is now in the lowest position.
- (k) Repeat Steps (d) to (j).
- (l) Turn the coupler so that the part of the coupler, which was previously attached to the horizontal tube, is now attached to the vertical tube. The tightening bolt or tightening wedge, attaching the coupler to the horizontal tube, now is in the uppermost position.
- (m) Repeat Steps (d) to (j).
- (n) Invert the coupler over so that the tightening bolt or tightening wedge referred to in Step (e) is in the lowest position.
- (o) Repeat Steps (d) to (j).

B6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of couplers tested.
- (b) The number of couplers that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4.
- (c) The test force applied to the scaffold tube.
- (d) For each coupler position, the amount of vertical plane rotation, in millimetres.
- (e) A reference to this test method, i.e. AS/NZS 1576.2, Appendix B.



NOTE: Coupler shown is swivel coupler.

DIMENSIONS IN MILLIMETRES

FIGURE B1 DISTORTION TEST APPARATUS FOR RIGHT-ANGLE AND SWIVEL COUPLERS

APPENDIX C
STRENGTH TEST FOR PUTLOG COUPLER
(Normative)

C1 SCOPE

This Appendix sets out the method for determining the strength of a putlog coupler on metal scaffold tube.

C2 PRINCIPLE

The coupler is fixed to two scaffold tubes at right angles. A predetermined force is applied to one of the tubes.

C3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure C1.
- (b) Scaffold tubes for which the coupler has been designed.
- (c) Torque wrench.
- (d) Timer.

C4 TEST COMPONENTS

New or as new hot-dip galvanized tubes with a wall thickness of not less than 4.0 mm shall be used. The mass of zinc per unit area shall be not less than 300 g/m². When tested, the tubes shall have a surface condition similar to that of a tube just after galvanizing.

C5 PROCEDURE

The procedure shall be as follows:

- (a) Fix the coupler on the scaffold tubes as shown in Figure C1.
- (b) For screw-tightened couplers, lubricate with light machine oil the thread of the bolt.
- (c) Tighten the coupler to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten the coupler to the requirements of Clause 3.2.3 and wait a minimum of 10 min before applying the test force.
- (e) Apply a vertical test force of 2.3 kN progressively over a period of 15 s and hold it for 1 min –0, +15 s.
- (f) Remove the force and record whether the putlog coupler had any visible gap between the tubes, or if the tube that was loaded can be moved by hand.

C6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of couplers tested.
- (b) The number of couplers that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4.
- (c) The test force applied to the scaffold tube.
- (d) The test result as recorded at Step (f) of Paragraph C5.
- (e) A reference to this test method, i.e. AS/NZS 1576.2, Appendix C.

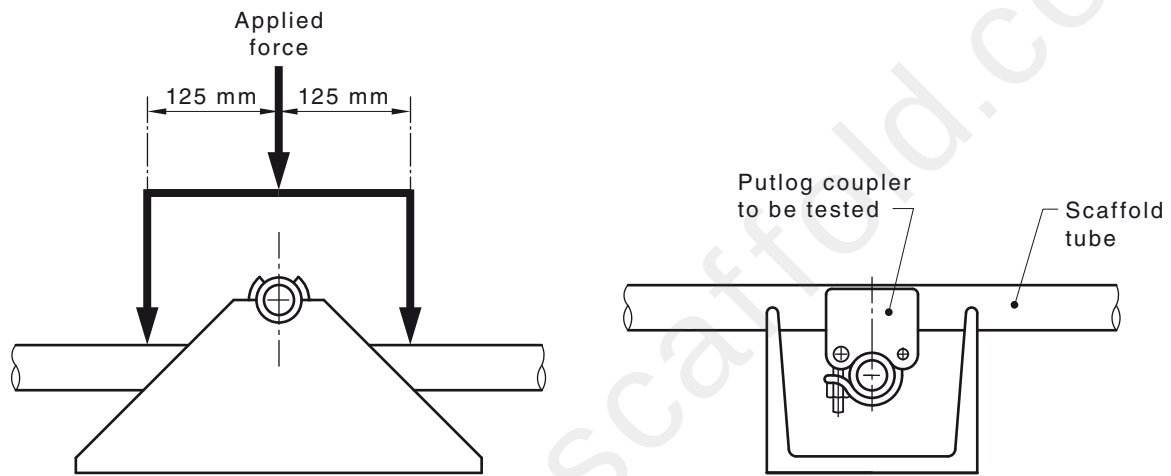


FIGURE C1 STRENGTH TEST APPARATUS FOR PUTLOG COUPLERS

APPENDIX D

SLIP TEST FOR PUTLOG COUPLER

(Normative)

D1 SCOPE

This Appendix sets out the method for determining the slip of a putlog coupler on metal scaffold tube.

D2 PRINCIPLE

The coupler is fixed to two scaffold tubes at right angles. A predetermined force is applied to one of the tubes. The amount of slippage of the coupler on the tube is measured.

D3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure D1. The frame shall be a stiff structure that does not deflect under test loads.
- (b) Scaffold tubes for which the coupler has been designed.
- (c) Torque wrench.

D4 TEST COMPONENTS

The test specimens (i.e. the putlog couplers) shall have been previously tested in accordance with Appendix C. When tested, the tubes shall have a surface condition similar to that of a new or as-new tube. Where the galvanized surface has been marked by the slippage of a coupler during a prior test, a fresh length of tube shall be used.

D5 PROCEDURE

The procedure shall be as follows:

- (a) Fix the coupler on new or as-new galvanized steel scaffold tubes as shown in Figure D1.
- (b) For screw-tightened couplers, lubricate the thread of the bolt with light machine oil.
- (c) Tighten the coupler to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten the coupler to the requirements of Clause 3.2.3.
- (e) Scribe a mark on the vertical tube immediately above the coupler.
- (f) Apply a vertical test force of 1.25 kN progressively over a period of 15 s and hold it for 1 min –0, +15 s.
- (g) Remove the force, scribe a new mark immediately above the coupler and record whether the coupler slipped on the tube and, if so, the amount of slippage in millimetres.

D6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- The number of couplers tested.
- The number of couplers that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4.
- The test force applied to the coupler.
- Whether the putlog tube slipped in the coupler and, if so, the amount of slippage in millimetres.
- A reference to this test method, i.e. AS/NZS 1576.2, Appendix D.

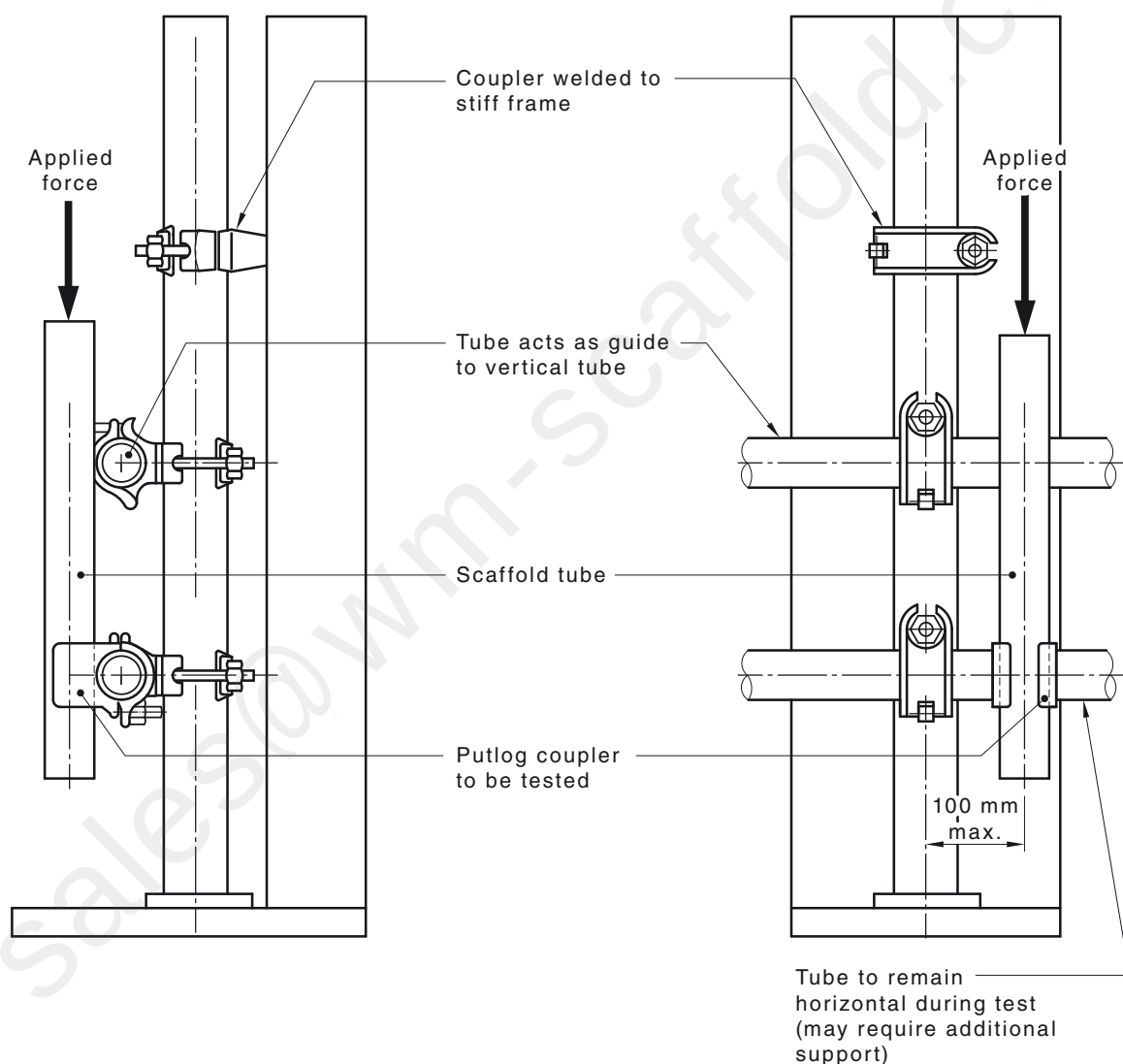


FIGURE D1 SLIP TEST APPARATUS FOR PUTLOG COUPLERS

APPENDIX E

BENDING LOAD CAPACITY OF END-TO-END COUPLER

(Normative)

E1 SCOPE

This Appendix sets out the method for determining the bending strength of internal and external end-to-end couplers when these are used to join metal scaffold tubes.

E2 PRINCIPLE

The coupler joins two tubes. The tubes are supported at a defined distance apart. A designated force is applied perpendicular to the weakest bending axis or plane of the coupler. A preliminary test shall be conducted to determine the weakest bending axis or plane. A predetermined force is applied to both tubes between the supporting points and the deflection of the coupler measured at the centre of the coupler.

E3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure E1.
- (b) Scaffold tubes for which the coupler has been designed.
- (c) Torque wrench for a screw-tightened coupler.
- (d) The tightening device for a wedge-tightened coupler in accordance with Clause 3.2.3(b).
- (e) Timer.

E4 TEST COMPONENTS

The test specimens shall be in accordance with Paragraph C4 of Appendix C.

E5 PROCEDURE

Internal and external types of end-to-end couplers shall be tested.

The procedure shall be as follows:

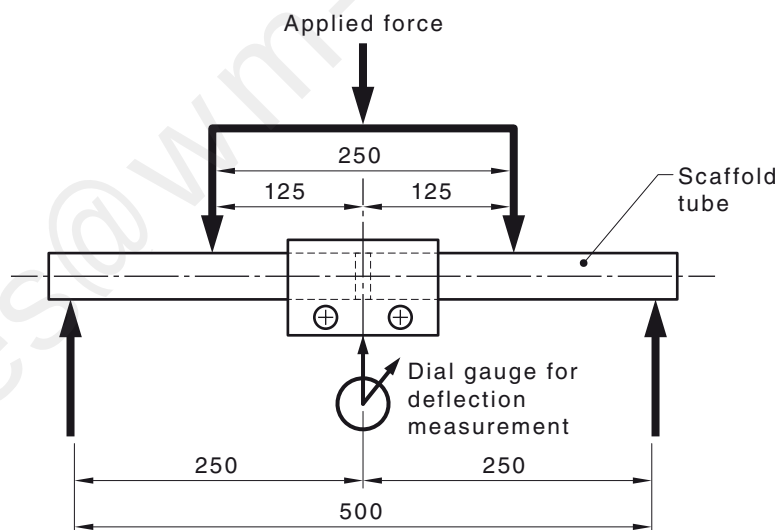
- (a) Make square cuts to the ends of the tubes that are enclosed in or around the coupler.
- (b) For screw-tightened couplers, lubricate with light machine oil the thread of the bolt.
- (c) Fix the coupler to both scaffold tubes ensuring that both tubes bear against the positive stop in the coupler. Nuts shall be free running on the bolt.
- (d) Tighten the coupler to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (e) Tighten the coupler to the requirements of Clause 3.2.3 and wait a minimum of 5 min before applying the test force.
- (f) Place the test assembly in the test rig so that the test force can be applied perpendicular to the weakest bending axis or plane of the coupler.

- (g) Subject the coupler to an initial vertical force of 1.0 kN progressively over a period of 15 s and hold it for a period of 1 min –0, +15 s (the initial force is part of the final force). Under this initial force, establish the datum for the deflection measurement at the joint between the tubes.
- (h) For an external type coupler, increase the force progressively to 15 kN over not less than 90 s, measuring the deflection for each 2 kN increase in the force.
- (i) For an internal type coupler, increase the force progressively to 7.5 kN over not less than 90 s, measuring the deflection for each 2 kN increase in the force.
- (j) Conclude the test when the measured deflection reaches 12 mm or the applied force reaches 15 kN for an external type coupler, or 7.5 kN for an internal type coupler, whichever occurs first.

E6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of couplers tested.
- (b) The number of couplers that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4.
- (c) The location of the weakest bending axis or plane of the coupler.
- (d) The test force applied.
- (e) The measured deflection, in millimetres.
- (f) A reference to this test method, i.e. AS/NZS 1576.2, Appendix E.



NOTES:

- 1 Coupler shown is an external sleeve coupler.
- 2 Distances between supports and load application points are clear distances.

DIMENSIONS IN MILLIMETRES

FIGURE E1 TEST APPARATUS FOR BENDING LOAD CAPACITY OF END-TO-END COUPLERS

APPENDIX F

SPLICING TEST FOR PARALLEL COUPLERS

(Normative)

F1 SCOPE

This Appendix sets out the method for determining the slip of two parallel couplers used to splice metal scaffold tubes.

F2 PRINCIPLE

Two parallel couplers are connected to two parallel scaffold tubes. A predetermined force is applied. Any slippage of the couplers on the tubes is observed.

F3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure F1.
- (b) Scaffold tubes for which the coupler has been designed.
- (c) Torque wrench for screw-tightened coupler.
- (d) The tightening device for a wedge-tightened coupler in accordance with Clause 3.2.3(b).
- (e) Timer.

F4 TEST COMPONENTS

When tested, the tubes shall have a surface condition similar to that of a new or as-new tube. Where the surface has been marked by slippage of a coupler during a prior test, a fresh length shall be used.

F5 PROCEDURE

The procedure shall be as follows:

- (a) Fix the couplers on the scaffold tubes as shown in Figure F1.
- (b) For screw-tightened couplers, lubricate the thread of the bolt with light machine oil.
- (c) Tighten the couplers to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten the couplers to the requirements of Clause 3.2.3 and wait a minimum of 10 min before applying the test force.
- (e) Subject the couplers to a vertical force of 6 kN progressively over a period of 15 s and hold it for 1 min –0, +15 s. The force shall be applied in the same manner as the test force (the force is to permit the coupler to bed in). All readings shall be taken after this force has been released.
- (f) Re-tighten the couplers to the requirements of Clause 3.2.3, and mark the couplers' position on all tubes.
- (g) Scribe a mark on the tubes immediately above both couplers.
- (h) Apply a vertical test force of 12.5 kN progressively over a period of 30 s and hold it for 1 min –0, +15 s.

- (i) Remove the force, scribe new marks on the tubes immediately above both couplers and record, in millimetres, the amount of slippage of the couplers on the tubes and any change in gap between the tubes. A measurable gap shall remain at conclusion of test (see Figure F1).

F6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- The number of couplers tested.
- The number of couplers that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4 and Paragraph F5.
- The test force applied to the scaffold tube.
- The amount of slippage of each coupler on each tube, in millimetres.
- A reference to this test method, i.e. AS/NZS 1576.2, Appendix F.

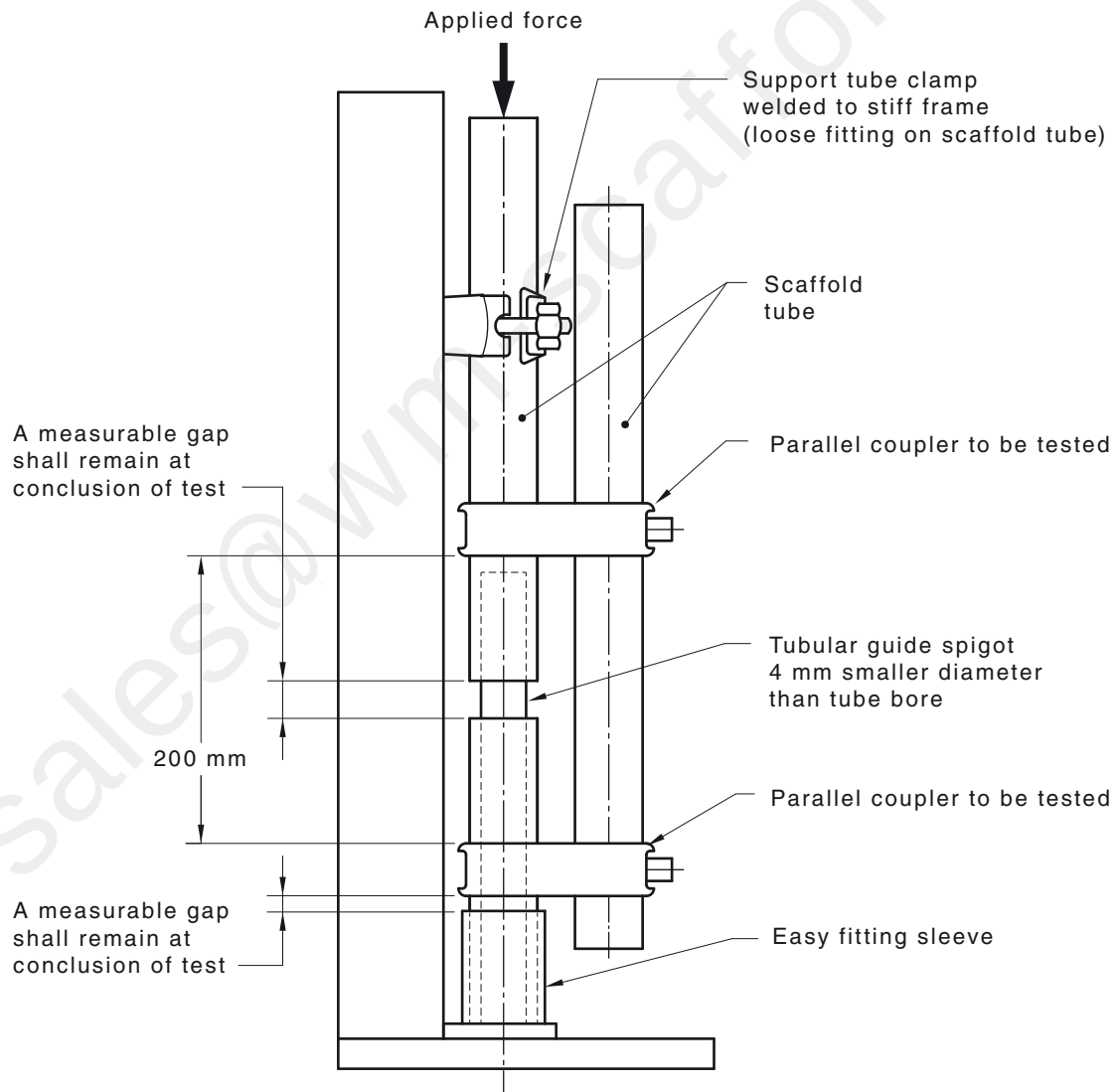


FIGURE F1 TEST APPARATUS FOR SPLICING TEST FOR PARALLEL COUPLERS

APPENDIX G

DETERMINATION OF SLIP RESISTANCE USING A CHECK COUPLER

(Normative)

G1 SCOPE

This Appendix sets out the method for determining the slip resistance of a right-angle clamping coupler on metal scaffold tube when backed up by a check coupler, where both couplers shall have satisfied the requirements of Appendix A.

G2 PRINCIPLE

The check coupler is fixed hard against the hinge pin end of the right-angle clamping coupler. The amount of load-producing slip is measured and is reduced by a factor of two to determine the working slip resistance of the clamping coupler when backed up by a check coupler. The check coupler may be a right-angle or a swivel coupler.

G3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure G1. The frame shall be a stiff structure that does not deflect under test loads.
- (b) Scaffold tubes for which the coupler has been designed.
- (c) Torque wrench for a screw-tightened coupler.
- (d) The tightening device for a wedge-tightened coupler in accordance with Clause 3.2.3(b).
- (e) Timer.

G4 TEST COMPONENTS

The test specimens shall be in accordance with Paragraph C4 of Appendix C.

G5 PROCEDURE

The procedure shall be as follows:

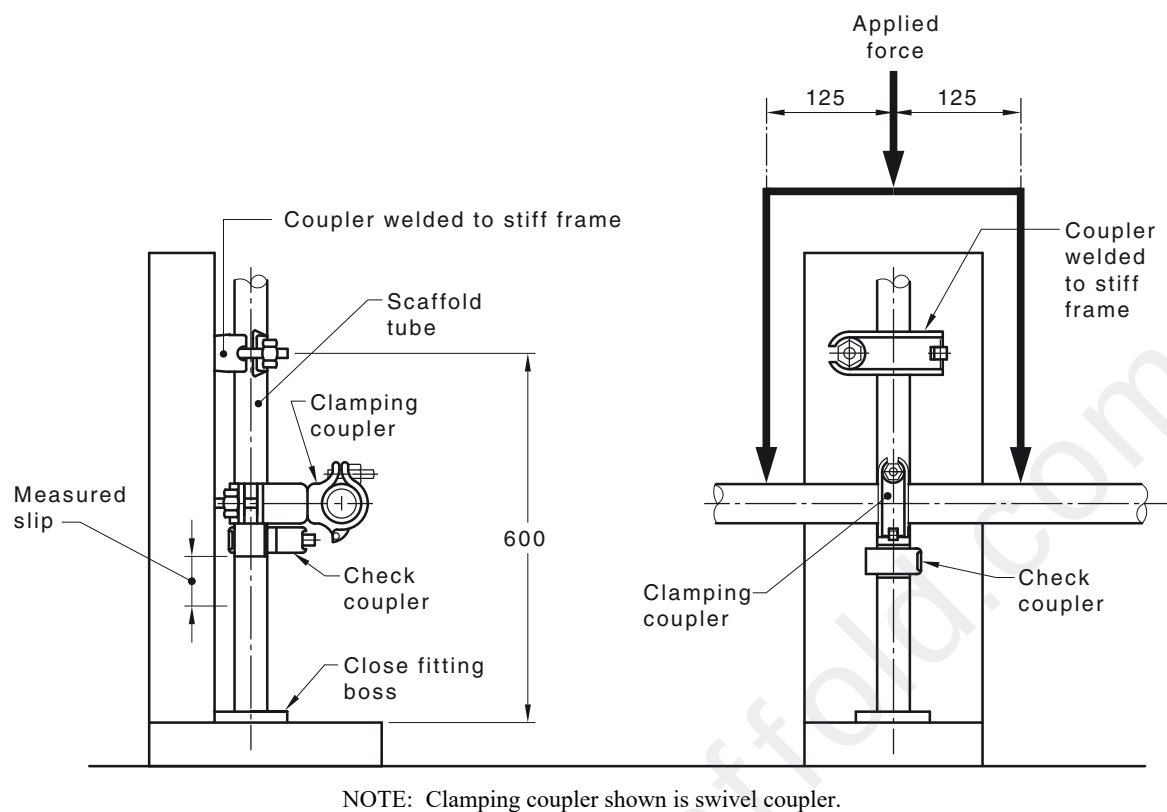
- (a) Fix the clamping coupler to the tubes as shown in Figure G1. Where the coupler incorporates a tightening bolt, it shall be in the uppermost position. Where the coupler uses a wedge for tightening, the wedge shall be in the orientation in which it was designed to be used. Fix the check coupler hard against the hinge pin end of the clamping right-angle coupler. Nuts shall be free running on the bolt.
- (b) For screw-tightened couplers, lubricate the thread of the bolt with light machine oil.
- (c) First tighten the clamping coupler and then the check coupler for conformity with the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten both couplers, in the same sequence as in Step (c), to the requirements of Clause 3.2.3 and wait a minimum of 10 min before applying the initial settling force in a vertical direction.
- (e) Subject the horizontal tube to a vertical force of 6 kN, applied progressively over a period of 15 s, and hold it for 1 min –0, +15 s. The force shall be applied in the same manner as the test force (the force is to permit the couplers to bed in). All readings shall be taken after this force has been released.

- (f) Re-tighten both couplers to the requirements of Clause 3.2.3.
- (g) Scribe a mark on the vertical tube immediately above the clamping coupler.
- (h) Apply in vertical direction a progressively increasing force. When slip commences, hold the force for 1 min –0, +15 s. Record this force.
- (i) Remove the force, scribe a new mark on the tube immediately above the clamping coupler and measure and record the amount of slippage, in millimetres.

G6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The type and number of clamping couplers tested.
- (b) The type and number of check couplers tested.
- (c) The torque applied to the screw-tightened clamping and check couplers.
- (d) The force applied to the scaffold tube at which the clamping coupler commenced to slip.
- (e) The amount of slippage, in millimetres.
- (f) The working slip resistance, in kilonewtons, of the clamping right-angle coupler when backed up by the check coupler, this being half the load at which slip commenced.
- (g) The number of couplers that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4 and Paragraph G5.
- (h) A reference to this test method, i.e. AS/NZS 1576.2, Appendix G.



DIMENSIONS IN MILLIMETRES

FIGURE G1 SLIP TEST APPARATUS FOR CHECK COUPLERS

APPENDIX H

VERTICAL LOAD TEST FOR ADJUSTABLE LEG, ADJUSTABLE BASEPLATE AND ADJUSTABLE SWIVEL BASEPLATE

(Normative)

H1 SCOPE

This Appendix sets out the method for determining the strength of a steel adjustable leg, aluminium and steel adjustable baseplates and aluminium and steel adjustable swivel baseplate in a scaffolding application. An adjustable leg is converted to an adjustable baseplate for testing purposes to gauge the effect of an eccentric load on the leg.

H2 PRINCIPLE

The adjustable leg or adjustable baseplate has one end fitted inside a scaffold tube with the maximum extension of thread shank exposed. A predetermined force is applied to the assembly. The strength of the adjustable leg or adjustable baseplate is measured as a function of the movability of the adjusting nut and straightness of the shank after the load is removed.

H3 STEEL ADJUSTABLE LEG, ADJUSTABLE BASEPLATE AND ADJUSTABLE SWIVEL BASEPLATE

H3.1 Apparatus

The following apparatus is required:

- (a) A device capable of applying a compressive load between two platens, as shown in Figure H1.
- (b) A 300 mm min. length of tube appropriate to the shank diameter with ends square cut to the tube axis. The tube shall be aluminium or steel depending on the end use of the adjustable baseplate.
- (c) A steel plate tapered to a slope of 1:10.
- (d) Timer.
- (e) A stop to prevent sliding of the swivel baseplate on the tapered plate.

The adjustable leg or adjustable baseplate shall be assembled in a vertical orientation (see Figure H1) with the upper portion of the shank fitted inside a scaffold tube and the adjusting nut set at the full extension. The upper portion of the shank shall extend not more than 150 mm in the scaffold tube.

For an adjustable swivel baseplate, the swivelling part of the baseplate, resting on the sloping base of the machine lower platen, shall follow the slope of the base. To prevent the swivelling part of the baseplate from sliding under the test load, the baseplate shall be restrained as appropriate.

H3.2 Procedure

The procedure shall be as follows:

- (a) Set the nut of the adjustable leg or adjustable baseplate to the maximum working extension of the shank below the nut. Check the straightness of the shank in three positions 45° apart and record any deviation.
- (b) Fit the tube over the adjustable leg or adjustable baseplate and place the assembly on the tapered plate between the testing machine platens.

- (c) Apply a force of 60 kN progressively over a period of 3 min and hold it for a period of 15 min –0, +15 s.
- (d) Remove the force.
- (e) Measure and record the straightness of the shank in the same positions as before the test.
- (f) Determine and record the ease with which the adjusting nut can be turned by hand for the full depth of the nut, downwards from the test position and returned upwards to the test position. If it is not possible to turn the nut, freely by hand, this shall be deemed as failure.
- (g) If the adjustable leg or adjustable baseplate passes this load test, re-test the adjustable leg or adjustable baseplate with an increasing load until failure (that is, the adjustable leg or adjustable baseplate is not able to support an increase in load or there is slip between the thread of the nut and the thread of the stem). This load shall be recorded as the ultimate failing load.

H3.3 Report

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of adjustable legs or adjustable baseplates tested.
- (b) The test force applied to the adjustable leg or adjustable baseplate assembly.
- (c) The change in straightness of the shank and the ease with which the adjusting nut can be turned by hand for the full depth of the nut upwards and downwards from the test position for the 60 kN test load.
- (d) The number of adjustable legs or adjustable baseplates that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4 and Paragraph H3.2.
- (e) The failure load for each component tested and mode of failure.
- (f) A determination of the WLL in accordance with Clause 4.2.1 or 4.2.2.
- (g) A reference to this test method, i.e. AS/NZS 1576.2, Appendix H.

H4 ALUMINIUM ADJUSTABLE BASEPLATE AND ADJUSTABLE SWIVEL BASEPLATE

H4.1 Apparatus

The following apparatus is required:

- (a) A device capable of applying a compressive load between two platens as shown in Figure H1.
- (b) A 300 mm min. length of tube appropriate to the shank diameter with ends square cut to the tube axis. The tube shall be aluminium or steel depending on the end use of the adjustable baseplate.
- (c) A steel plate tapered to a slope of 1:10.
- (d) Timer.
- (e) A stop to prevent sliding of the swivel baseplate on the tapered plate.

The adjustable baseplate shall have the upper portion of the shank fitted inside a scaffold tube assembled in a vertical orientation (see Figure H1) and the adjusting nut set to the maximum extension.

For an adjustable swivel baseplate, the swivelling part of the baseplate, resting on the sloping base of the machine lower platen, shall follow the slope of the base. To prevent the swivelling part of the baseplate from sliding under the test load, the baseplate shall be restrained as appropriate.

H4.2 Procedure

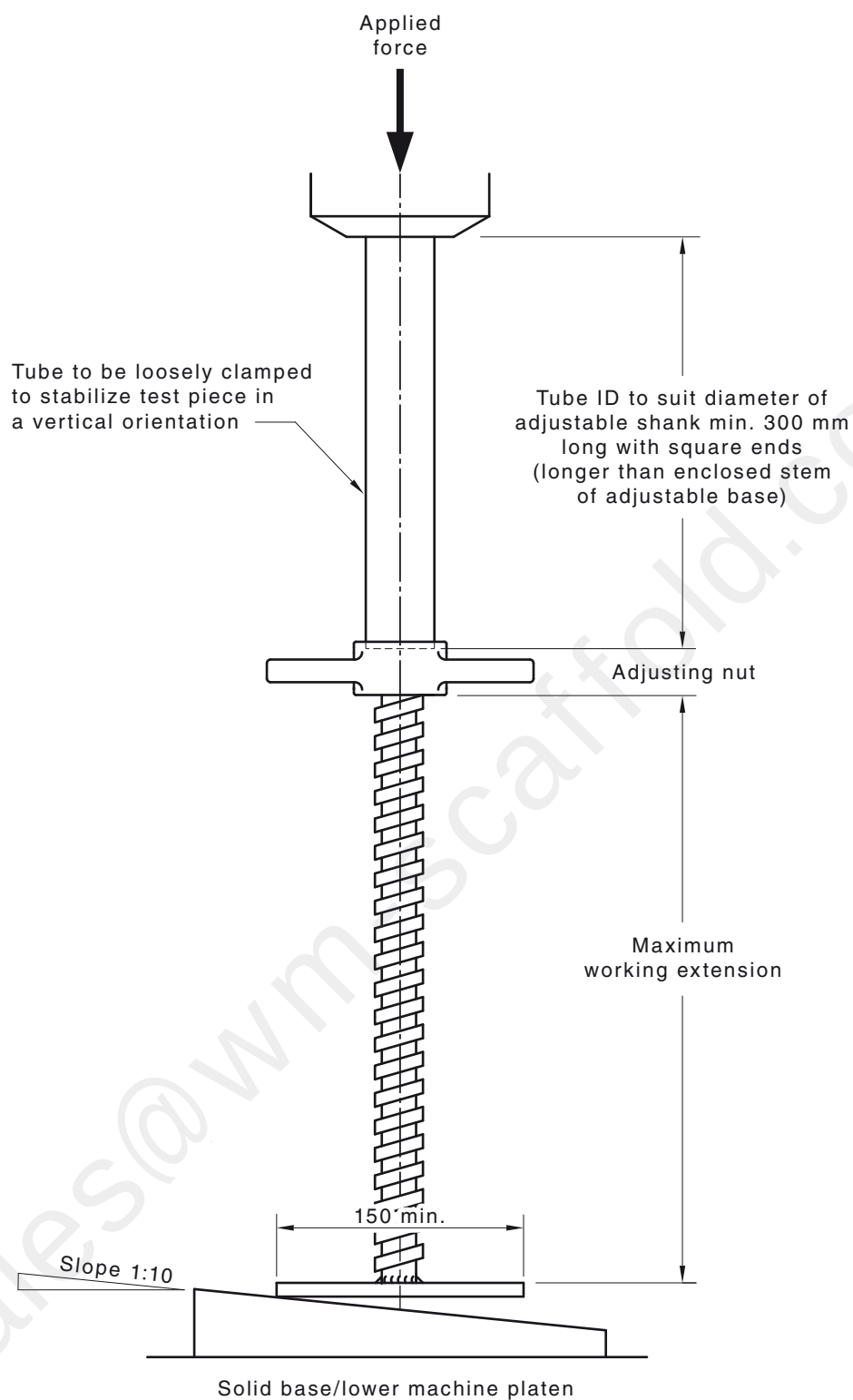
The procedure shall be as follows:

- (a) Set the nut of the adjustable leg or baseplate to the maximum working extension of the shank below the nut. Check the straightness of the shank in three positions 45° apart and record any deviation.
- (b) Fit the tube over the adjustable leg or baseplate and place the assembly on the tapered plate between the testing machine platens.
- (c) Apply a vertical force of 30 kN progressively over a period of 3 min and hold it for a period of 5 min –0, +15 s.
- (d) Remove the force.
- (e) Measure and record the straightness of the shank in the same positions as before the test.
- (f) Determine and record the ease with which the adjusting nut can be turned by hand for the full depth of the nut, downwards from the test position and returned upwards to the test position. If it is not possible to turn the nut, freely by hand, this is to be regarded as failure.
- (g) If the adjustable base passes this load test, re-test the adjustable base with an increasing load until failure (that is, the adjustable base is not able to support an increase in load or there is slip between the thread of the nut and the thread of the stem). This load is to be recorded as the ultimate failing load.

H4.3 Report

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of adjustable baseplates tested.
- (b) The test force applied to the adjustable baseplate assembly.
- (c) The change in straightness of the shank and the ease with which the adjusting nut can be turned by hand for the full depth of the nut upwards and downwards from the test position for the 30 kN test load.
- (d) The number of adjustable baseplates that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4 and Paragraph H4.2.
- (e) The failure load for each component tested and mode of failure.
- (f) A determination of the WLL in accordance with Clause 4.2.3 or 4.2.4.
- (g) A reference to this test method, i.e. AS/NZS 1576.2, Appendix H.



NOTE: Adjustable bases to be tested with their normal baseplate.

DIMENSIONS IN MILLIMETRES

FIGURE H1 TEST APPARATUS FOR VERTICAL LOAD TEST ON ADJUSTABLE LEG OR BASEPLATE

APPENDIX I
VERTICAL LOAD TEST FOR A PLAIN PINTLE CASTOR
AND ADJUSTABLE CASTOR AT NO EXTENSION

(Normative)

I1 SCOPE

This Appendix sets out the method for determining the strength of a plain pintle castor, including the castor component of an adjustable castor at no extension, to support the vertical load imposed by a scaffold supported on a horizontal level surface.

I2 PRINCIPLE

A predetermined force is applied to the castor and the permanent set, if any, is measured and reported.

I3 APPARATUS

A device capable of applying a compressive load between two platens and retaining the castor during loading.

I4 PROCEDURE

The procedure shall be as follows:

- (a) Fit the castor in a device that will grip only the pintle in a manner similar to fixing the castor in the end of an aluminium or steel standard. Apply the brake to the castor and fit the assembly between the testing machine platens either vertically or horizontally.
- (b) Apply progressively over a period of not less than 60 s a vertical test force of 1.5 times the rated load and hold it for 15 min –0, +15 s, during which time the castor shall not develop a permanent set.
- (c) Remove the force and measure and record any permanent set.
- (d) If there is no permanent set, apply progressively over a period of not less than 2 min a vertical test force of three times the rated load and hold it for 15 min –0, +15 s, during which time the castor shall not collapse.
- (e) Remove the test force.
- (f) Inspect the castor and note whether the castor collapsed or not.

I5 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of castors tested.
- (b) The WLL marked on the castors and the test force applied to the castor.
- (c) The amount of permanent set.
- (d) Record the number of castors that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4 and Paragraph I4, and the modes of failure.
- (e) A reference to this test method, i.e. AS/NZS 1576.2, Appendix I.

APPENDIX J

VERTICAL LOAD TEST ON AN ADJUSTABLE CASTOR

(Normative)

J1 SCOPE

This Appendix sets out the method for determining the strength of an adjustable castor to support the vertical load imposed by a scaffold when the supporting surface is not horizontal.

NOTE: AS/NZS 1576.1 permits the use of adjustable castors to support the scaffold load where the gradient of the supporting surface does not exceed 5° from the horizontal.

J2 PRINCIPLE

The adjustable castor has one end fitted inside the scaffold tube of the internal diameter to suit the castor leg. The maximum extension of the threaded shank shall be exposed. A predetermined force is applied to the assembly. The strength of the adjustable castor is measured as a function of the movability of the adjusting nut and the change in straightness of the adjustable shank.

J3 APPARATUS

The following apparatus is required:

- (a) A device capable of applying a compressive load between two platens, as shown in Figure J1.
- (b) A 300 mm min. length of tube appropriate to the castor shank diameter with ends square cut to the tube axis. The tube shall be aluminium or steel depending on the end use of the adjustable castor.
- (c) A steel plate tapered to a slope 1:10.
- (d) A timer.

J4 PROCEDURE

The procedure shall be as follows:

- (a) Set the nut of the adjustable shank to the maximum extension. Check the straightness of the shank in three positions 45° apart and record deviation.
- (b) Fit the tube over the shank. Apply the break to the castor and fit the assembly on the tapered plate between the testing machine platens. The adjustable castor shall be angled down to slope as show in Figure J1. Note and record if the tube is of aluminium or steel.
- (c) Apply progressively over a period of not less than 2 min a vertical test force of two times the rated load and hold it for 5 min $\pm$ 15 s, during which time the adjustable castor shall not collapse.
- (d) Remove the test force.
- (e) Inspect the castor and note whether any components separated or failed and whether the adjustable castor collapsed.
- (f) Measure and record the straightness of the adjustable shank in the same positions as before the test.

- (g) Determine and record the ease with which the adjusting nut can be turned by hand downwards from the test position and returned upwards to the test position. If it is not possible to turn the nut freely by hand, this shall be recorded as failure.

J5 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of adjustable castors tested.
- (b) The WLL marked on the castors and the test force applied.
- (c) The material of the tube specified as noted in Step J4(b).
- (d) Whether or not any components failed or separated and whether or not the adjustable castor collapsed.
- (e) The change in straightness of the shank of each adjustable castor tested.
- (f) The number of the adjustable castors that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4 and Paragraph J4.
- (g) Record the failure force, the mode of failure, any permanent set or out of straightness of the shank for each adjustable castor tested.
- (h) A reference to this test method, i.e. AS/NZS 1576.2, Appendix J.

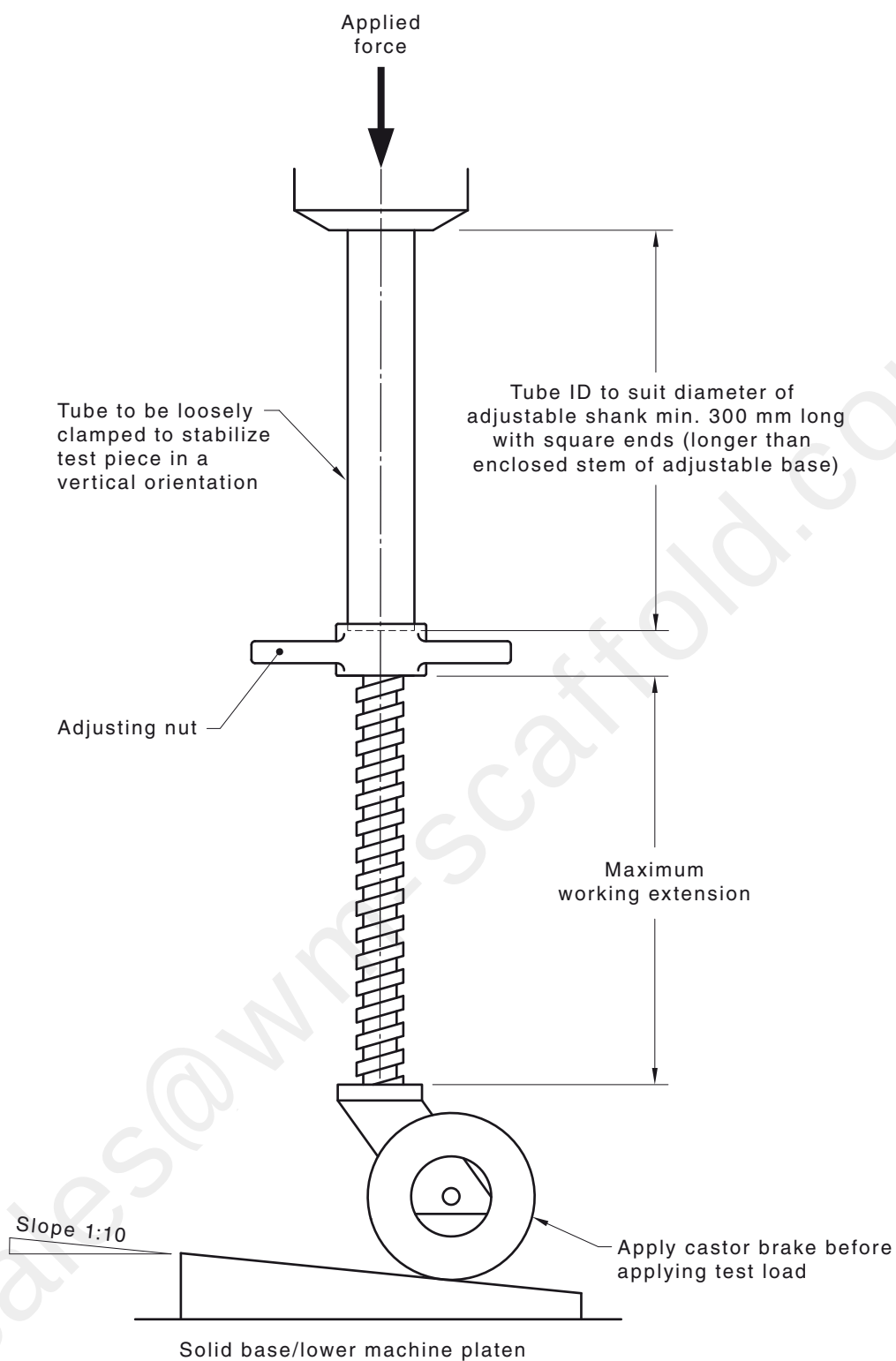


FIGURE J1 TEST APPARATUS FOR VERTICAL LOAD TEST ON CASTOR

APPENDIX K
TEST FOR PUTLOG BLADE
(Normative)

K1 SCOPE

This Appendix sets out the method for determining the strength of a putlog blade.

K2 PRINCIPLE

The putlog blade is attached to a putlog tube. A predetermined force is applied to the tube adjacent to the fitting. The permanent set, if any, is measured and reported.

K3 APPARATUS

The following apparatus is required:

- (a) Support arrangement as shown in Figure K1.
- (b) Scaffold tube for which the putlog blade has been designed.
- (c) Torque wrench for a screw-tightened putlog blade.
- (d) The specified tightening device for a wedge-tightened putlog blade.
- (e) Timer.

K4 TEST COMPONENTS

The test specimens shall be in accordance with Paragraph C4, Appendix C.

K5 PROCEDURE

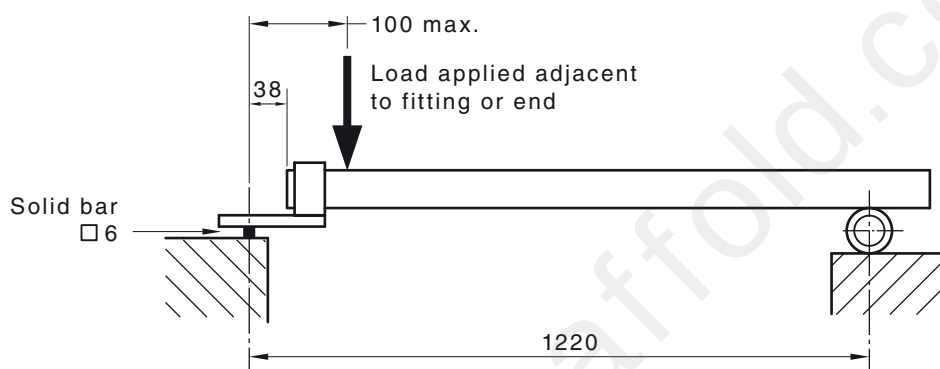
The procedure shall be as follows:

- (a) Fix the putlog blade to the scaffold tube, as shown in Figure K1.
- (b) For screw-tightened couplers, lubricate with light machine oil the thread of the bolt.
- (c) Tighten the putlog blade to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten the putlog blade to the requirements of Clause 3.2.3 and wait a minimum of 10 min prior to applying the test force.
- (e) Support the tube and putlog blade as shown in Position A in Figure K1.
- (f) Apply to the tube, at a distance of 100 mm from the centre of the 6 mm solid bar, a vertical force of 4.5 kN progressively over a period of 15 s and hold it for 1 min –0, +15 s.
- (g) Remove the force and measure and record any permanent set.
- (h) If there is no permanent set, support the tube and putlog blade as shown in Position B in Figure K1.
- (i) Repeat Steps (f) and (g).

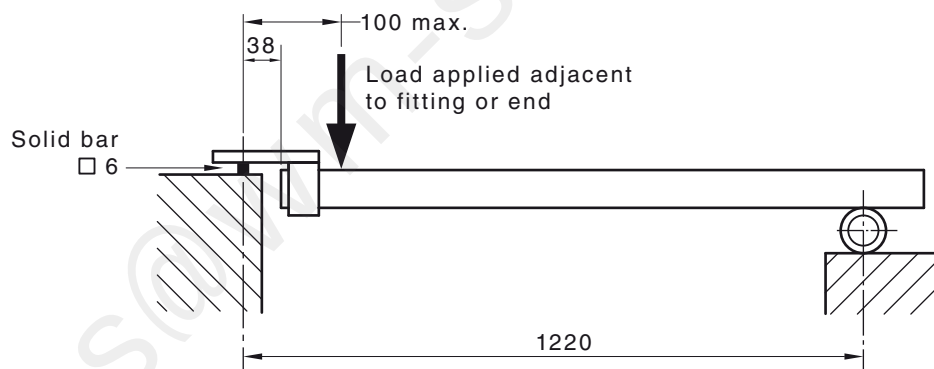
K6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of putlog blades tested.
- (b) The number of putlog blades that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4.
- (c) The test force applied to the tube.
- (d) The amount of permanent set for Position A and for Position B.
- (e) A reference to this test method, i.e. AS/NZS 1576.2, Appendix K.



(a) Position A



(b) Position B

DIMENSIONS IN MILLIMETRES

FIGURE K1 TEST APPARATUS FOR PUTLOG BLADE

APPENDIX L
STRENGTH TEST FOR FLANGE CLAMP
(Normative)

L1 SCOPE

This Appendix sets out the method for determining the strength of a pair of rigid or swivel flange clamps.

L2 PRINCIPLE

Two flange clamps are used to attach a scaffold tube to a tapered flange or universal beam. A predetermined tensile force is applied and permanent set, if any, is measured and reported.

L3 APPARATUS

The following apparatus is required:

- (a) A device capable of applying a tensile load as shown in Figure L1.
- (b) Test apparatus as shown in Figure L1.
- (c) Timer.
- (d) Torque wrench.

L4 PROCEDURE

The procedure shall be as follows:

- (a) Fix the flange clamps in the test apparatus, as shown in Figure L1.
- (b) Tighten the clamp to the requirements of Clause 3.2.3.
- (c) Subject the flange clamps to a tensile force of 30 kN, applied progressively over a period of 90 s and hold it for 1 min –0, +15 s.
- (d) Remove the force and record whether the flange clamps showed any permanent set.

L5 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of flange clamps tested.
- (b) The number of flange clamps that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4.
- (c) The test force applied to the flange clamp.
- (d) A reference to this test method, i.e. AS/NZS 1576.2, Appendix L.

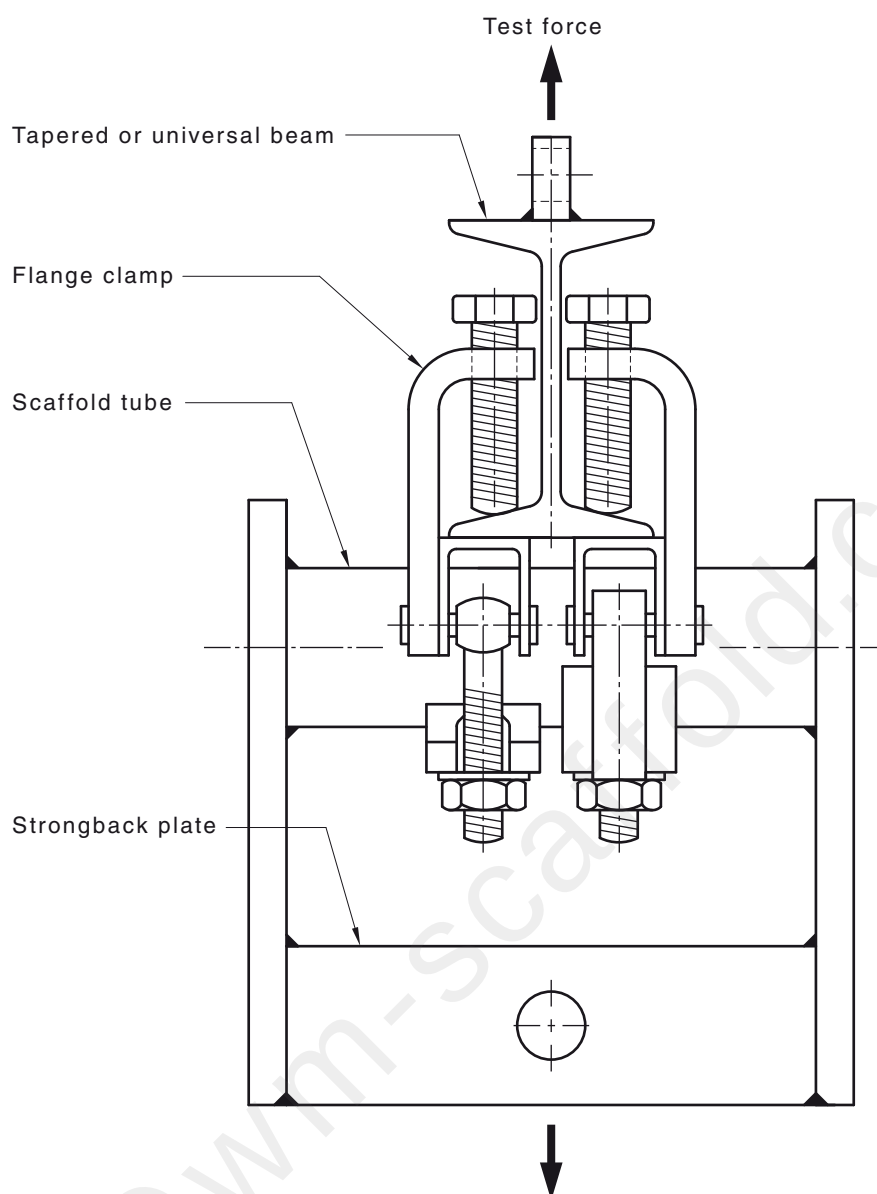


FIGURE L1 TEST APPARATUS FOR RIGID AND SWIVEL FLANGE CLAMP

APPENDIX M

SLIP TEST THROUGH RIGID AND SWIVEL FLANGE CLAMPS

(Normative)

M1 SCOPE

This Appendix sets out the method for determining the slip of a metal scaffold tube through a pair of rigid or swivel flange clamps.

M2 PRINCIPLE

Two flange clamps, rigid or swivel, are used to attach a scaffold tube to a tapered flange or universal beam. A predetermined axial force is applied to the tube. The amount of slippage of the tube relative to the beam flange is measured.

M3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure M1.
- (b) Scaffold tube for which the flange clamps have been designed.
- (c) Torque wrench.
- (d) Timer.

M4 TEST COMPONENTS

The test specimens shall be in accordance with Paragraph C4, Appendix C. Where the galvanized surface has been marked by slippage of a coupler during a prior test, a fresh length of tube shall be used. The surface of the beam shall be clean and free of mill scale, rust or paint.

M5 PROCEDURE

The procedure shall be as follows:

- (a) Fix a pair of flange clamps, rigid or swivel, to hold the scaffold tube to the flange of a steel beam. For swivel flange clamps, the tube shall be fixed in the same orientation as for the rigid flange clamps.
- (b) For screw-tightened couplers, lubricate with light machine oil the thread of the bolt.
- (c) Tighten the flange clamps to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten the flange clamps to the requirements of Clause 3.2.3 and wait a minimum of 10 min before applying the initial settling in force.
- (e) Subject the tube to an axial force of 6 kN, applied progressively over 15 s and hold it for 1 min –0, +15 s (the force is to permit the flange clamps to bed in). All readings shall be taken after this force has been released.
- (f) Re-tighten the flange clamps to the requirements of Clause 3.2.3.
- (g) Scribe a mark on the tube in line with one edge of the beam flange.
- (h) Subject the tube to an axial force of 12.5 kN, applied progressively over 30 s and hold it for 1 min –0, +15 s.

- (i) Remove the force and scribe a new mark on the tube in line with the same edge of the beam flange as in Step (f). Measure and read the amount of slip.

M6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of flange clamps tested.
- (b) The number of flange clamps that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4 and Paragraph M5.
- (c) The test force applied to the scaffold tube.
- (d) The amount of slippage of the tube, in millimetres.
- (e) Whether the flange clamps passed or failed the test.
- (f) A reference to this test method, i.e. AS/NZS 1576.2, Appendix M.

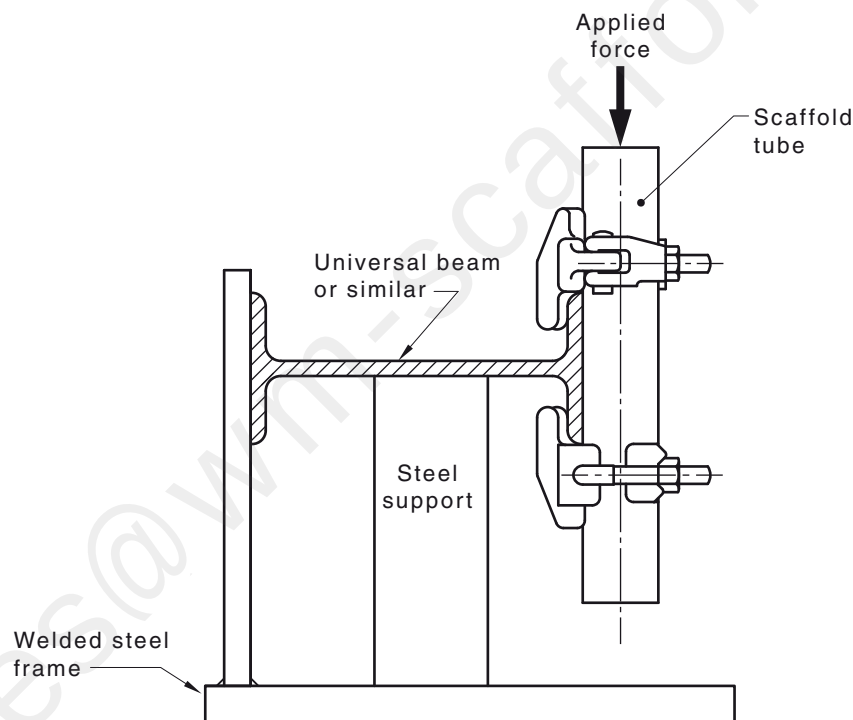


FIGURE M1 TEST APPARATUS FOR SLIP OF TUBE THROUGH THE FLANGE CLAMPS

APPENDIX N

SLIP TEST OF RIGID AND SWIVEL FLANGE CLAMPS ALONG A FLANGE

(Normative)

N1 SCOPE

This Appendix sets out the method for determining the amount of slippage of flange clamps along the flange of the beam.

N2 PRINCIPLE

Two flange clamps, rigid or swivel, are used to attach a metal scaffold tube to a tapered flange or universal beam. A predetermined transverse force is applied to the tube. The amount of slippage of both flange clamps along the beam flange is measured.

N3 APPARATUS

The following apparatus is required:

- (a) Test apparatus as shown in Figure N1.
- (b) Scaffold tube for which the flange clamps have been designed.
- (c) Torque wrench.
- (d) Timer.

N4 TEST COMPONENTS

The test specimens shall be in accordance with Paragraph C4, Appendix C. The surface of the beam shall be clean and free of mill scale, rust or paint.

N5 PROCEDURE

The procedure shall be as follows:

- (a) Fix a pair of clamps, rigid or swivel, to hold the tube to a flange of a steel beam. For swivel clamps, the tube shall be fixed in the same orientation as for the rigid clamps.
- (b) For screw-tightened couplers, lubricate with light machine oil the thread of the bolt.
- (c) Tighten the clamp bolts to the requirements of Clause 3.2.3 five times prior to commencing the test.
- (d) Tighten the flange clamps to the requirements of Clause 3.2.3 and wait a minimum of 10 min before applying the initial settling in force.
- (e) Subject the tube to a transverse force, parallel to the flange, of 6 kN progressively over 15 s and hold it for 1 min –0, +15 s (the force is to permit the flange clamps to bed in). All readings shall be taken after this force has been released.
- (f) Re-tighten the flange clamps to the requirements of Clause 3.2.3.
- (g) Scribe marks on the beam flange in line with the top of the flange clamps.
- (h) Subject the tube to a transverse force, parallel to the flange, of 12.5 kN progressively over 30 s and hold it for 1 min –0, +15 s.
- (i) Remove the force and scribe new marks on the beam flange in line with the top of the flange clamps. Measure and record the amount of slippage of each clamp along the beam flange.

N6 REPORT

In addition to the requirements of Appendix O, the report shall include the following information:

- (a) The number of flange clamps tested.
- (b) The number of flange clamps that passed or failed the acceptance criteria when assessed in accordance with the performance requirements of Section 4 and Paragraph N5.
- (c) The test force applied to the scaffold tube.
- (d) The amount of slippage for each flange clamp, in millimetres.
- (e) Whether the flange clamps passed or failed the test.
- (f) A reference to this test method, i.e. AS/NZS 1576.2, Appendix N.

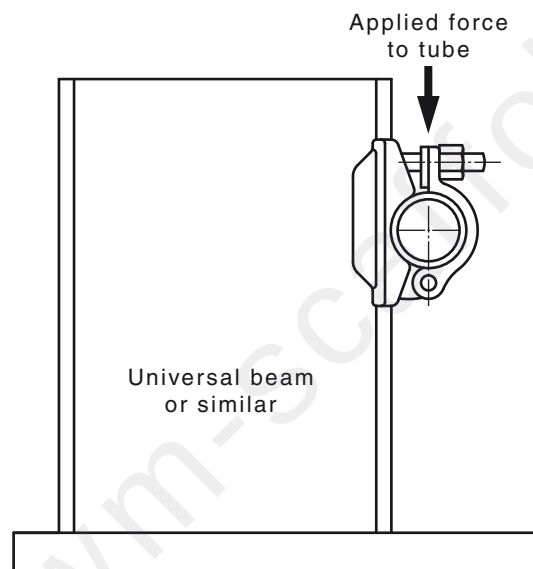


FIGURE N1 TEST APPARATUS FOR SLIP OF FLANGE CLAMP ALONG A FLANGE

APPENDIX O
TEST REPORTS
(Normative)

Reports of tests carried out on scaffold couplers and accessories shall include the following information:

- (a) Type of coupler or accessory tested.
- (b) Manufacturer's or supplier's identification of the coupler or accessory and batch reference number.
- (c) Detailed or numbered drawing or photograph of the coupler or accessory.
- (d) Detailed description, drawing or photograph of the test apparatus.
- (e) Location of testing facility.
- (f) Date of test.
- (g) Name, position and qualification of the person responsible for the test.
- (h) The outcome of the test required by the Appendix regulating the test.
- (i) A reference to the test method (e.g. Appendix A of AS/NZS 1576.2).

The report shall be signed and dated by the person responsible for the test. When more than one set of tests is included in one report, the report as a whole need only be signed, not each set of tests.

APPENDIX P
PRODUCT CONFORMITY
(Informative)

P1 SCOPE

This Appendix sets out the following different means by which conformity with this Standard can be demonstrated by the manufacturer or supplier:

- (a) Evaluation by means of statistical sampling.
- (b) The use of a product certification scheme.
- (c) Assurance using the acceptability of the supplier's quality system.
- (d) Other such means proposed by the manufacturer or supplier and acceptable to the customer.

P2 STATISTICAL SAMPLING

Statistical sampling is a procedure which enables decisions to be made about the quality of batches of items after inspecting or testing only a portion of those items. This procedure will only be valid if the sampling plan has been determined on a statistical basis and the following requirements are met:

- (a) The sample needs to be drawn randomly from a population of product of known history. The history needs to enable verification that the product was made from known materials at essentially the same time, by essentially the same processes and under essentially the same system of control.
- (b) For each different situation, a suitable sampling plan needs to be defined. A sampling plan for one manufacturer of given capability and product throughput may not be relevant to another manufacturer producing the same items.

In order for statistical sampling to be meaningful to the customer, the manufacturer or supplier needs to demonstrate how the above conditions have been satisfied. Sampling and the establishment of a sampling plan should be carried out in accordance with AS 1199.1, guidance to which is given in AS 1199.0.

P3 PRODUCT CERTIFICATION

The purpose of product certification is to provide independent assurance of the claim by the manufacturer that products conform to the stated Standard.

Guidance on certification schemes is given in ISO/IEC TR 17026 (and SANZ/SAA HB18.28 for New Zealand).

A certification scheme serves to indicate that the products consistently conform to the requirements of the Standard in conjunction with type testing and a sampling regime. The manufacturer's methods for the control of production may also be the subject of a certification scheme.

P4 SUPPLIER'S QUALITY MANAGEMENT SYSTEM

Where the manufacturer or supplier can demonstrate an audited and registered quality management system conforming to the requirements of the appropriate or stipulated Australian or International Standard for a supplier's quality management system or systems, this may provide the necessary confidence that the specified requirements will be met. The quality assurance requirements need to be agreed between the customer and supplier and should include a quality or inspection and test plan to ensure product conformity.

Information on establishing a quality management system is set out in AS/NZS ISO 9001 and AS/NZS ISO 9004.

P5 OTHER MEANS OF ASSESSMENT

If the above methods are considered inappropriate, determination of conformity with the requirements of this Standard may be assessed from the results of testing coupled with the manufacturer's guarantee of product conformance.

Irrespective of acceptable quality levels (AQLs) or test frequencies, the responsibility remains with the manufacturer or supplier to supply products that conform to the full requirements of the Standard.

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- 1199.0 Part 0: Introduction to the ISO 2859 attribute sampling system
- 1199.1 Part 1: Sampling schemes indexed by acceptance quality limit (AQL) for lot-by-lot inspection

AS/NZS ISO

- 9001 Quality management systems—Requirements
- 9004 Quality management systems—Guidelines for performance improvements

SANZ/SAA

- HB 18 Guidelines for third-party certification and accreditation
- HB 18.28 Guide 28 General rules for a model third-party certification scheme for products

ISO/IEC TR

- 17026 Conformity assessment—Example of a certification scheme for tangible products

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